



Ideology-driven social media opinions and capital markets: Evidence from polarizing boycotts

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ABSTRACT

Investors increasingly turn to social media for information, even though many sites are saturated with ideological opinions. This paper examines whether ideology-driven social media opinions affect stock market responses to polarizing firm news, using boycotts as the empirical setting. On average, polarizing boycotts are associated with a 1% (2.3%) drop in equity value over the seven (60) trading days after gaining social media traction. Consistent with ideology-driven opinions affecting market responses, the immediate price reaction is more negative when users ideologically aligned with boycotters dominate social media, particularly when their opinions are visible and financially relevant. A mediation analysis suggests that this association operates indirectly through overall social media sentiment toward the boycotted firms. Additionally, trading volume and daily return volatility following boycotts are positively associated with user ideological diversity. Together, these results suggest that ideology-driven social media opinions could influence how investors respond to polarizing firm news.

1. Introduction

Social media have become popular venues for the expression of ideological opinions (Pew Research Center, 2024). Studies show that these opinions can shape political behaviors (e.g., Karell et al., 2023; Müller and Schwarz, 2023), but little is known about their effects on financial decision-making. This question is important, as the academic and practical literatures document that investors acquire and trade on information from social media, such as X and StockTwits (e.g., Cookson and Niessner, 2020; Jia et al., 2020; CFA Institute, 2024).¹ In this study, I examine whether political ideology influences investment-focused social media users' opinions about polarizing firm news and, if so, whether ideology-driven opinions affect market reaction to the news.² I define polarizing news as news that generates substantial disagreement among individuals with different ideological beliefs.

My study involves three empirical challenges: first, identifying polarizing firm news; second, identifying social media users' ideologies; and third, determining these users' views of firms. To address the first challenge, I use boycotts over firms' perceived stances or actions on ideologically polarizing issues ("polarizing boycotts" hereafter) as empirical proxies for polarizing news. In the U.S., many issues, such as climate change, gun policy, and LGBTQ rights, are highly politicized. Therefore, firms can be boycotted when they take stances on polarizing issues or when their actions are interpreted as supporting liberal or conservative positions. For instance, in 2023,

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¹ Twitter was rebranded as "X" in July 2023. I only use the term "Twitter" to refer to data, tools, and services that preceded the rebranding, such as the Twitter Academic API and the Twitter RoBERTa model.

² For brevity, I use "social media users" to refer to investment-focused social media users. This term only refers to social media users who reference firms' cashtags in their posts; those who mention firm-related hashtags (e.g., #boycottcocacola) without referencing cashtags are excluded.

Anheuser-Busch and Target faced boycotts from conservatives over their marketing campaigns celebrating Pride Month.³ On the other hand, Home Depot unintentionally became entangled in controversy when a former executive donated \$1.75 million to controversial Republican Senate candidate Herschel Walker, sparking boycotts by liberals.⁴ Given the close connection between polarizing issues and political ideology, I conjecture that social media users who share the boycotters' ideology and who are therefore more likely to oppose the boycotted firms' perceived stances, express more negative sentiment toward these firms. These ideology-driven opinions, in turn, can have implications for stock market responses. Specifically, I predict more negative price reaction to polarizing boycotts when users ideologically aligned with boycotters dominate social media and greater investor disagreement when the ideological beliefs of social media users show more diversity.

I use X trending topics to construct a sample of 125 boycotts resulting from perceived corporate stances on polarizing issues, as identified by the USC Polarization Index (USC Center for Public Relations, 2022). X trending topics are viral topics prominently displayed on x.com and the mobile app and frequently covered by the media, raising the likelihood that social media users will follow the news. For each boycott in my sample, I classify its prevailing ideology by reading the posts and referencing contemporaneous news articles, ending up with 80 (64%) boycotts consistent with liberal ideologies and 45 (36%) consistent with conservative ideologies.⁵

To address the second challenge of classifying social media users' ideologies, I follow an established methodology in political science that detects social media users' ideologies based on the accounts they follow (e.g., Barberá, 2015; Halberstam and Knight, 2016). Specifically, I compile a list of 2,072 partisan X accounts and identify which accounts each user follows. I classify a user as a liberal (conservative) if she follows more liberal (conservative) accounts than conservative (liberal) ones.

Turning to the third challenge of determining users' views of boycotted firms, I analyze the 93,447 posts containing the boycotted firms' cashtags—the dollar sign followed by a ticker symbol—posted when boycotts trended on X.⁶ Including a cashtag in posts indicates awareness of and interest in its stock, so users who do so likely have interest or expertise in investment. To gauge social media sentiment toward boycotted companies, I instruct ChatGPT (GPT-4 model) to classify post sentiment toward boycotted firms (i.e., entity-level sentiment) as positive, neutral, or negative. Comparing manual and model classification results within a random sample of 500 posts, GPT-4 achieves a 76% (72%) accuracy (F1) score, outperforming Twitter RoBERTa-base, the Loughran-McDonald dictionary (Loughran and McDonald, 2011), and FinBERT.⁷

I begin my empirical analyses by describing boycott causes and firm responses. Sixty-six percent of the 125 boycotts result from firm actions, particularly product or operational decisions, human resources decisions, and political contributions or advocacy. Former or current employee actions or other factors, such as politicians' remarks about firms or firms' inaction on polarizing issues, trigger the other 34%. In terms of firm responses to boycotts, the most common type is staying silent (39%), followed by explaining (33%), reversing the controversial policy (16%), or doubling down (12%). Firms that choose to respond do so quickly, taking on average just under three days after the boycott starts trending.

I next examine the social media activity surrounding the boycotts. Firms targeted by polarizing boycotts attract online attention, with an average of 748 cashtag posts over a two-day event window, which tend to express negative sentiment. For reference, the average viral earnings announcement identified by Campbell et al. (2023) is associated with 279 cashtag posts over the same window length. On average, conservative-initiated boycotts related to political events attract the most engagement, while liberal-initiated boycotts related to racial equality generate the most negative social media sentiment. Social media users posting during boycotts are ideologically diverse: 33% are classified as liberals, 44% as conservatives, and the remaining 23% as nonpartisans. Compared to liberal and conservative accounts, nonpartisan ones post far more frequently and are more likely to display bot-like behaviors, suggesting that many could be automated accounts.

To confirm that social media users' ideologies affect their views of boycotted firms, I test whether social media users ideologically aligned with boycotters express more negative sentiment toward the boycotted firm, and vice versa. On the one hand, ideologies could influence online investment opinions by leading users with different beliefs to selectively acquire or interpret information through partisan lenses or by affecting their senses of identity congruence with boycotted firms. On the other hand, users may distance themselves from political motivations to appear as credible financial commentators. Consistent with ideologies affecting investment opinions, I find that social media users express more negative sentiment toward boycotted firms on X when they align ideologically with the boycotters. This result is robust to including boycott fixed effects and controlling for other user characteristics, suggesting that it is unlikely driven by other firm or user characteristics.

Turning to the main research question of whether ideology-driven social media opinions affect market reaction to polarizing news, I start by describing the market reaction to polarizing boycotts. First, I find that polarizing boycotts are associated with an approximate

³ See <https://www.axios.com/2023/06/16/corporate-america-pride-backlash-stocks>.

⁴ See <https://www.newsweek.com/home-depot-clears-rumors-donations-herschel-walkers-campaign-1750033>.

⁵ See Appendix IA2 in the online appendix for detailed boycott ideology classification procedures.

⁶ I limit my attention to posts containing cashtags because clicking on cashtags allows users to easily access all content related to a firm, reducing information processing costs relative to posts that reference firms using non-standardized names (e.g., brand or product variations). Consistent with this, studies extensively show that investors trade on information in posts with cashtags (e.g., Bartov et al., 2018; Jia et al., 2020; Campbell et al., 2023), whereas the decision relevance of posts without cashtags to investors is not well established. In addition, posts with cashtags can be easily queried via the Twitter Academic API, whereas those referencing non-standardized firm or brand names are substantially harder to collect and contain more noise.

⁷ Accuracy is the proportion of all classifications that are correct. The F1 score measures an algorithm's precision and recall. A score of 1 indicates perfect precision and recall, and a score of 0 indicates zero precision or recall.

1% (2.3%) drop in equity value over seven (60) trading days after they start trending on X. Consistent with ideology-driven opinions having market consequences, immediate market reaction is more negative when a greater proportion of posts are by users ideologically aligned with the boycotters, a pattern that also exists for volume-based retail order imbalance. To strengthen the interpretation of the price response result, I conduct a path analysis and find that this negative association operates indirectly through overall social media sentiment toward boycotted firms. In cross-sectional analyses focusing on posts more likely to influence trading, I find that this negative association is concentrated among posts that have greater visibility or discuss the financial performance of the boycotted firms. These results suggest that ideology-driven social media opinions could exacerbate the negative market response to polarizing boycotts.

In addition to directional responses, I also examine whether heterogeneity in social media users' ideologies is associated with investor disagreement about boycott news. I find that social media users' ideological diversity, calculated as the inverse Herfindahl index of their ideological leanings, is positively associated with abnormal trading volume following polarizing boycotts. I also find a similar but weaker association between user ideological diversity and post-boycott abnormal daily return volatility. These results suggest that greater ideological diversity among social media users is positively associated with investor disagreement.

While these findings are consistent with ideology-driven social media opinions affecting capital markets, I acknowledge several caveats. First, because of the cross-sectional design, other factors could confound the relation between ideology-driven online discussions and market outcomes. To mitigate this concern, I control for boycotted firm visibility, concurrent news sentiment, and each boycott's polarizing nature in all market-based tests. Cross-sectional tests show that this association is concentrated among posts more likely to be viewed and deemed financially relevant by investors. Further, a path analysis suggests that the association operates through social media sentiment toward boycotted firms, though it does not rule out the existence of omitted confounding variables. Second, reverse causality is another concern, as market reaction can affect social media opinions rather than the other way around. I find that the results are robust to excluding posts that reference boycotted firms' stock performance. Nevertheless, I cannot rule out reverse causality concerns. Third, the small sample size may limit the power of my tests. However, trending boycotts are highly visible, and I document significant effects across multiple tests. Finally, because this paper focuses on polarizing news, I cannot speak to whether ideology-driven social media opinions affect investor responses to apolitical news. However, the findings are likely to generalize to other polarizing events, such as firms' statements or silence on socio-political issues, where there is increasing pressure to take a stance.⁸

Despite these caveats, this paper contributes to both research and practice. First, it adds to the literature on how social media shape investor information processing (Lee et al., 2015; Jia et al., 2020; Campbell et al., 2023) by showing how the ideologies of social media users can influence investors' responses to politically salient firm news. A related study by Cookson et al. (2020) examines whether divergent ideologies of StockTwits users are associated with investor disagreement during the COVID-19 pandemic. My study takes a different perspective by examining polarizing boycotts targeting individual firms. These boycotts can be consistent with liberal or conservative ideologies and encompass a multitude of issues, including climate change, gun policy, and LGBTQ rights. My study also broadens the research on political leanings of financial information intermediaries (Baloria and Heese, 2018; Rees and Twedt, 2022; Goldman et al., 2024) by extending it to social media users, who have increasingly become a trusted source of financial information, particularly among younger investors (CFA Institute, 2024).

Second, this paper contributes to the literature on the benefits and costs of social media attention to firms. While research highlights benefits, such as improved visibility and liquidity (Blankespoor et al., 2014; Rakowski et al., 2021), recent work also points to downsides, including distorted price discovery, lower price efficiency, and greater toxicity (Jia et al., 2020; Campbell et al., 2023; Blankespoor et al., 2025). Complementing this literature, I find that ideology-driven social media opinions can amplify negative market responses to polarizing corporate crises and that opinions from users with conflicting ideologies are positively associated with daily return volatility. Given growing concern among practitioners and regulators that social media may destabilize markets (PwC, 2017; Board of Governors of the Federal Reserve System, 2021), these findings suggest that politically charged online discourse could shape market outcomes.

2. Background and hypothesis development

2.1. Polarizing boycotts

The political science literature defines boycotts of businesses as consumers' deliberate avoidance of certain products with the aim of changing corporate practices (Stolle and Micheletti, 2013). Boycotts can happen when firms take stances on polarizing issues or when they are perceived to support one side of a debate.

As an example of the former, in the spring of 2023, companies like Anheuser-Busch InBev (AB InBev) and Target introduced marketing campaigns in support of the LGBTQ community, with Bud Light partnering with transgender influencer Dylan Mulvaney and Target releasing its Pride Month product line. In response, conservative public figures protested, aiming to "make 'pride' toxic for

⁸ For example, in January 2026, local communities and pension funds pressured Minnesota-based firms, such as Target, to take a stance against ICE agent actions in Minneapolis. See <https://www.cbsnews.com/news/target-ceo-michael-fiddelke-ice-minneapolis-minnesota/>.

brands.”⁹ Subsequently, both AB InBev and Target experienced declining stock prices and analyst downgrades.¹⁰ As an example of the latter, during Herschel Walker's campaign for the U.S. Senate in 2022, a Twitter user alleged that Home Depot had donated to his campaign and called on liberals to boycott the company. Home Depot denied the claim, clarifying that an executive who had left the firm two decades earlier had donated.¹¹ This highlights the fact that boycotts can arise from inaccurate perceptions.

Social media today serve as commons for the discussion of boycotts. As retired U.S. Supreme Court Justice Anthony Kennedy observed, “[t]hese websites [Facebook and Twitter] can provide perhaps the most powerful mechanisms available to a private citizen to make his or her voice heard.”¹² Viral posts can elevate a boycott from a niche political action to a global phenomenon. Moreover, because a wide audience, including investors, often sees social media discussions, these discussions can have implications for the stock market.

2.2. Hypothesis development

An active community of users on X frequently discusses financial news and investment strategies. These users can include investors, financial analysts, journalists, and others interested in disseminating financial information. Faced with polarizing boycotts, they may develop divergent opinions about the boycotted firms.

I conjecture that ideology can influence how investment-focused social media users interpret boycott-related news for financial or nonfinancial reasons. From a financial perspective, recent studies suggest that, faced with the same event, individuals with different ideologies may acquire different information or interpret the same information differently, leading to divergent investment opinions (e.g., Cookson et al., 2020; Meeuwis et al., 2022; Kempf et al., 2023). Therefore, social media users ideologically aligned with boycotters may be more likely to consume news from or adopt the views of like-minded sources. As a result, they may be more likely to express negative opinions toward the boycotted firms. In contrast, those ideologically opposed to boycotters are likely to conclude the opposite. From a nonfinancial perspective, social identity theory argues that shared identity increases the extent to which individuals identify with a group (Tajfel and Turner, 1986) and vice versa. During boycotts, social media users ideologically aligned with boycotters may experience lower identity congruence with the boycotted firm, prompting them to voice more negative opinions online.

Academic and practical literatures document that investors acquire and trade on information from social media, such as X and StockTwits (e.g., Cookson and Niessner, 2020; Rakowski et al., 2021; CFA Institute, 2024). For example, Cookson and Niessner (2020), Jia et al. (2020), and Campbell et al. (2023) find that investment-related discussions on social media significantly shape market reactions to events, such as earnings announcements and merger rumors. In addition, a 2023 Brunswick survey of institutional investors found that over 90 percent of respondents said they used social media to investigate investment decisions.¹³ Building on this literature, I predict that ideology-driven social media opinions can influence how the stock market responds to polarizing firm news, such as boycotts.

I acknowledge tension in this prediction. To begin, research finds mixed evidence on whether ideology-driven boycotts can meaningfully harm firm performance, as boycotts are often counteracted by “buycotts,” in which consumers intentionally buy products to support boycotted firms. While Chavis and Leslie (2009) find that the boycott of French wine in response to France's opposition to the Iraq War led to a sustained decline in wine sales, recent studies (Jin et al., 2025; Wang and Lu, 2022; Liaukonytė et al., 2023) find that “buycotts” can offset the negative effects of boycotts. Thus, it is unclear whether investors will respond to boycott news. Second, if ideological fervor drives investment opinions expressed toward boycotted companies on social media and those opinions lack economic substance, investors may disregard them as irrelevant to their trading. Therefore, it is an open question whether ideology-driven opinions on social media affect the stock market's response to boycotts.

To the extent that online opinions do inform trading decisions, though, I expect that the market reaction to polarizing boycotts will be more negative when a larger share of social media posts about the boycotted firm come from users ideologically aligned with boycotters, compared to when a smaller share of posts come from these users. Thus, I make the following prediction regarding post-boycott stock returns:

H1. The price response to an ideologically polarizing boycott is more negative when a larger share of posts about the boycotted firm comes from users ideologically aligned with the boycotters.

Excess return volatility is another market outcome that can be costly for firms and their shareholders. Theory suggests that, when investors hold heterogeneous beliefs, elevated disagreement leads to both higher trading volume and greater return volatility (Scheinkman and Xiong, 2003; Banerjee and Kremer, 2010). If conflicting ideologies among social media users prompt divergent investment opinions and if investors incorporate this disagreement into their trading, then social media discourse from more ideologically diverse users may be associated with higher trading volume and volatility than discourse from less diverse users. This leads to my second prediction:

⁹ See <https://x.com/MattWalshBlog/status/1661333191951613952>.

¹⁰ See <https://www.cnbc.com/2023/05/10/hbsc-downgrades-ab-inbev-as-it-deals-with-a-bud-light-crisis.html>; <https://www.cnbc.com/2023/06/01/jpmorgan-downgrades-target-as-concerns-for-the-retail-giant-mount.html>.

¹¹ See <https://x.com/HomeDepot/status/1578439312454324224>.

¹² *Packingham v. North Carolina*, 582 U.S. 98 (2017).

¹³ See <https://www.brunswickgroup.com/app/uploads/Digital-Investor-Survey-Two-Page-Summary.pdf>.

H2. Following an ideologically polarizing boycott, abnormal trading volume and abnormal realized volatility increase with the diversity of social media users' ideological beliefs.

3. Data and sample construction

To identify social media boycotts, I start from daily top 50 trending topics on X archived by www.trendcalendar.com and check whether each represents a boycott of public companies based on the matching procedure detailed in Appendix IA1 of the online appendix. I identify 256 trending boycotts targeting 92 public companies in the CRSP database from April 2016 through October 2022.

I then manually classify the boycotts into several categories based on the nature of the issues and tag their prevailing ideologies as liberal or conservative. (See detailed procedure in Appendix IA2 of the online appendix.) Out of this sample, I retain the 130 boycotts pertaining to corporate stances or actions regarding polarizing issues identified by the USC Polarization Index ([USC Center for Public Relations, 2022](https://usc.edu/polarization)), as I expect ideology to play a major role in how these boycotts are perceived and not the ones pertaining to non-polarizing issues. The issue categories are abortion, COVID-19, climate change, domestic/international politics, LGBTQ rights, gun policy, and racial equality/policing policy.¹⁴ I also exclude boycotts targeting foreign ADRs, because their prices are primarily determined by the prices of underlying shares in their respective home markets. This further reduces the number of boycotts to 125. See Table 1 for sample selection procedure and Appendix B for a complete list of the boycotts in the final sample.

For each boycott, I use the Twitter Academic API to download posts containing boycotted firms' cashtags in the days after these boycotts trended. I end up with 135,943 posts by 59,934 unique users. From this initial sample, I exclude posts outside the [0, +1] day window following when the boycotts trend. I then manually inspect the data to further exclude spam posts that reference target firms' cashtags incidentally.¹⁵ I also exclude posts missing control variables. My final sample consists of 93,447 posts with control variable data, stemming from 125 boycotts targeting 58 firms. These posts are generated by 45,349 unique users.

A potential concern is that posts containing cashtags may have been made not by investment-focused users but by people participating in the boycott. To reduce this concern, Table IA1 Panel A of the online appendix presents the most common words in the profiles of users in my sample, revealing a strong investment focus (e.g., "stocks," "crypto," "options," "investing"). Panel B contrasts the most frequent words in my sample (posts containing cashtags posted around boycotts) with those in posts containing boycott hashtags posted in the same period.¹⁶ The results indicate that, while my sample posts feature investment terms, generic posts around boycotts do not. This evidence suggests that the users in my sample are likely to have an investment focus.

To explore the relation between ideology-driven social media opinions and market reaction to boycotts, I also obtain stock market variables from CRSP and TAQ, along with financial performance variables from Compustat, analyst following data from I/B/E/S, institutional ownership data from Thomson Reuters, and traditional news media data from RavenPack.

4. Research design and descriptive statistics

4.1. Measurement of social media user ideology

Research in political science shows that the political ideology of social media users can be determined by analyzing the political accounts they follow. In a pioneering study, Barberá (2015) estimates Twitter users' political ideology based on the partisan accounts they follow. To validate this approach, he links a sample of Twitter accounts to real-world voters and campaign users, finding a strong correlation between Twitter-based ideology and party registration and likewise for campaign contribution. Applying the same method to classify Twitter user ideology, Halberstam and Knight (2016) show that users who follow mostly liberal accounts post favorably about Democrats and unfavorably about Republicans, and vice versa, further validating the classification method.

Drawing on this approach, I estimate social media user ideology from the partisan accounts they follow. Compared to classifying user ideology based on explicit partisan language in post histories (Cookson et al., 2020), this method has the advantage of directly capturing the potential partisan nature of users' information sets. It can also identify the beliefs of users who avoid using partisan language to appear as credible financial commentators. To begin, I collect the X accounts of U.S. federal and state and local politicians in office as of January 2023 from the following sources: ProPublica's Politiwoops project (<https://projects.propublica.org/politwoops/>), the UCSD Library website (<https://ucsd.libguides.com/congresstwitter>), the U.S. House of Representatives Press Gallery (<https://pressgallery.house.gov/member-data/members-official-twitter-handles>), and a list of cabinet members' accounts compiled by @TwitterGov (<https://x.com/i/lists/88345660>). Because these sources do not include accounts belonging to politically biased media organizations, pundits, or former politicians, I further collect political accounts from the datasets shared by Wojcieszak et al. (2022) and Green (2018) and manually compile an additional 39 liberal pundit accounts and 39 conservative pundit accounts from various online sources. After this step, there are 1036 liberal X accounts and 899 conservative X accounts. To ensure equal numbers of liberal

¹⁴ The USC Polarization Index further classifies political issues into subcategories, such as immigration, healthcare reform, Jan. 6 Capitol attack, Russia's attack on Ukraine, etc. For simplicity, I combine these issues into one category and label it "domestic/international politics."

¹⁵ Specifically, the posts I manually filtered out contain Hyatt's cashtag \$H, as \$H is sometimes used as part of a math formula or an intentional misspelling of "SH."

¹⁶ I was only able to collect posts containing boycott-related hashtags for 93 (74%) of the boycotts in the sample before the Twitter Academic API was shut down in June 2023. However, there is no reason to believe the most frequent words in posts about these boycotts do not represent the overall sample.

Table 1
Sample construction.

Panel A: Firm-Level Sample Selection		
	Number of boycotts	Number of unique firms
Trending boycotts targeting publicly traded companies from April 2016 through October 2022	256	92
<i>Exclude:</i> boycotts resulting from non-ideologically polarizing issues or whose boycotter ideology is ambiguous	(125)	(29)
<i>Exclude:</i> firms whose cashtags were never discussed on X	(1)	(1)
<i>Exclude:</i> foreign ADRs	(5)	(4)
Initial sample	125	58
Panel B: Post-Level Sample Selection		
	Number of posts	Number of unique users
Initial sample	135,943	59,934
<i>Exclude:</i> posts made outside the [0, +1] day window	(39,731)	(13,889)
<i>Exclude:</i> posts that are spams	(147)	(139)
<i>Exclude:</i> posts missing control variables	(2618)	(557)
Final sample	93,447	45,349

Notes: This table describes the sample selection procedure. Boycotts in the following categories are not ideologically polarizing according to the USC Polarization Index and therefore excluded: Business Events, Corporate Misconduct, Employee Rights, Gender Equality/Sexual Misconduct, Popular Culture Controversy, Privacy Breaches/Data Leaks, Product Safety/Customer Complaints, and Unfounded Rumors.

and conservative accounts, I obtain the 137 most highly followed conservative X accounts that are followed by two prominent conservative organizations (@Heritage and @NRA), ending up with 1036 liberal X accounts and 1036 conservative X accounts. Finally, I match these 2072 political accounts to each user's X following list and compute the number of liberal and conservative accounts each user follows. A user is classified as a liberal if she follows more liberal accounts than conservative ones, as a conservative if the reverse is true, or as a nonpartisan if she follows no political accounts or an equal number of liberal and conservative accounts.

Next, I assess the validity of this measure of ideology using two approaches. First, following [Cookson and Niessner's \(2020\)](#) method of using distinct words to confirm meaningful differences between categories, Panel A of [Table 2](#) lists the 15 most frequently used words in each user group's profiles that do not appear in the profiles of other groups. The top words are clearly divided along party lines: liberals often use terms like “voteblue” and “bidenharris,” while conservatives often use terms like “ultramaga” and “drain-theswamp.” Second, following [Halberstam and Knight \(2016\)](#), I correlate my ideology measure with 2020 presidential election results at the state level using users' self-disclosed location data.¹⁷ I find that, across states, the share of users classified as liberal (conservative) under my method and the share of 2020 Biden (Trump) voters are highly correlated, with a coefficient of 0.78 (0.79) significant at the 0.01 level. Figure IA1 in the online appendix further shows that, when plotting each state's share of users classified as liberals (conservatives) against its share of 2020 Biden (Trump) voters, most states align closely with the 45-degree line. These results suggest that my measure of ideology reasonably captures users' political leanings.

4.2. Measurement of social media sentiment toward the boycotted firm

To proxy for social media sentiment toward boycotted firms, I measure the sentiment expressed toward boycotted firms in each post. Although the widely used Twitter RoBERTa-base model generates high-quality classifications of post-level sentiment ([Loureiro et al., 2022](#)), several features of the posts in my sample complicate the task. First, these posts often reference multiple stocks and other entities and express conflicting sentiment toward them. Thus, it is necessary to isolate the sentiment expressed toward specific firms mentioned in a post. Second, it is often challenging for sentiment classifiers to detect subtle sentiments, such as sarcasm, which prevail in social media posts. For example, the following post subtly criticizes Pepsi: “what's the word for greenwashing but for ‘empowering women’? \$PEP @PepsiCo.”¹⁸ This post does not direct any overtly negative words at Pepsi, yet it conveys the author's discontent. Finally, many posts contain words that convey positive or negative sentiment only in an investment context (e.g., “long,” “short,” “call,” and “put”), but sentiment classifiers trained on a general social media corpus, like Twitter RoBERTa-base, tend to label them as neutral words.

To tackle these challenges, I use ChatGPT (GPT-4) to assess the sentiment in posts directed at boycotted firms. Released by OpenAI in November 2022, ChatGPT is a generative large language model (GLLM) applicable to many natural language processing tasks. [de Kok \(2025\)](#) points out that GLLMs can be better adapted to complex tasks than other textual analysis methods, and they can be more consistent and efficient than research assistants. In particular, with its proven capabilities in advanced NLP tasks ([OpenAI, 2023](#)), GPT-4 can handle the sentiment analyses required for this study. I construct a prompt that instructs GPT-4 to classify a post's sentiment toward a boycotted firm as positive, neutral, or negative. See [Appendix D](#) Panel A for the prompt and [Appendix E](#) for example

¹⁷ In my sample, 14,477 (32%) of the 45,349 users disclose their states of residency in profiles. State-by-state results from the 2020 U.S. presidential election are obtained from <https://www.presidency.ucsb.edu/statistics/elections/2020>.

¹⁸ See https://x.com/Bruno_J_Navarro/status/1486346090253279236.

Table 2

Descriptive statistics of users and ideologically polarizing boycotts.

Panel A: Most Common Unique Words in User Profiles by Political Ideology						
Ideology	Most Common Unique Words in User Profile					
Liberal (N = 15,235)	voteblue (77), bidenharris (44), standwithukraine (32), bluewave (31), khive (16), gqp (16), strongertogether (15), magats (15), npr (15), goodtrouble (15), stillwithher (14), racists (13), justicematters (12), indivisible (12), businessinsider (12)					
Conservative (N = 19,745)	ultramaga (63), pureblood (54), parler (48), gab (37), lgbfjb (20), desantis (19), lgb (18), trudeaumustgo (17), bluelivesmatter (16), endthefed (15), draintheswamp (14), freedoms (14), populist (13), saveamerica (12), miners (12)					
Nonpartisan (N = 10,369)	owler (9), alertchart (7), tranding (7), wiz (6), trialists (5), batter (5), furu (5), recovered (4), suhwu (4), trapmusic (4), cuttachasebeats (4), nftcollector (4), mymetatrader (4), crowdsale (4), wallstreetsignals (4)					
Panel B: Number of Boycotts by Issue Category and Boycotter Ideology						
Category	Number of boycotts with liberal boycotters		Number of boycotts with conservative boycotters	Total		
Abortion	3		1	4		
COVID-19	11		9	20		
Climate change	2		0	2		
Domestic/international politics	46		17	63		
Gun policy	3		3	6		
LGBTQ rights	3		8	11		
Racial equality/policing policy	12		7	19		
Total	80		45	125		
Panel C: Summary Statistics on Events That Triggered Boycotts						
Category	Sub-category		Number of Boycotts			
Firm action (n=82)	Firm's product/operational decision		47			
	Firm's human resource decision/policy		15			
	Firm's political contribution/advocacy		11			
	Firm's marketing/PR decision		8			
	Firm's disclosure		1			
Employee action (n=26)	Current executive statement/action		12			
	Current employee statement/action		9			
	Former executive statement/action		4			
	Former employee statement/action		1			
Other (n=17)	Non-firm action		8			
	Firm inaction		4			
	Subsidiary/franchisee action		5			
Total			125			
Panel D: Types of Firm Response to Polarizing Boycotts						
Response Type	Number (%) of Boycotts		Examples			
Silence	49 (39%)		N/A			
Explanation	41 (33%)		Meta's response to August 2022 boycott			
Policy Reversal	20 (16%)		Delta's response to May 2022 boycott			
Doubling Down	15 (12%)		Nike's response to July 2019 boycott			
Panel E: Number of Posts and Average User Sentiment						
Issue category	Liberal boycotts		Conservative boycotts		All boycotts	
	Avg. # of posts	Avg. post sentiment	Avg. # of posts	Avg. post sentiment	Avg. # of posts	Avg. post sentiment
Abortion	1248	0.32	54	0.06	950	0.31
COVID-19	299	0.03	825	-0.04	536	-0.02
Climate change	398	-0.14	-	-	398	-0.14
Domestic/international politics	467	-0.08	1823	-0.21	833	-0.16
Gun policy	779	0.02	194	0.10	486	0.04
LGBTQ rights	1670	0.10	745	-0.11	997	-0.01
Racial equality/policing policy	469	-0.22	879	-0.11	620	-0.16
Total	528	-0.03	1137	-0.16	748	-0.10
Panel F: Means and Medians of User Characteristics by Political Ideology						
	Liberals (N = 15,235)		Conservatives (N = 19,745)		Nonpartisans (N = 10,369)	
CountLiberalFollowing	53.28 (16)		5.31 (1)		0.35 (0)	
CountConservativeFollowing	6.04 (2)		63.82 (22)		0.35 (0)	
SentTowardsFirm	-0.13 (0)		-0.27 (0)		0.06 (0)	
UserVerified	0.04 (0)		0.01 (0)		0.01 (0)	
AccountAge	6.84 (7.34)		5.56 (5.06)		3.66 (1.83)	
FollowerCount	10,289.96 (640)		3773.56 (414)		3813.27 (138)	

(continued on next page)

Table 2 (continued)

Panel F: Means and Medians of User Characteristics by Political Ideology			
	Liberals (N = 15,235)	Conservatives (N = 19,745)	Nonpartisans (N = 10,369)
<i>FollowingCount</i>	2381.44 (1028)	1775.48 (681)	620.09 (137)
<i>PostCount</i>	70,794.82 (18,480.8)	42,376.73 (12,390)	24,706.66 (4001)
<i>FinCommentExp</i>	472.17 (11)	458.40 (18)	2361.61 (147)
<i>LikelyBot</i>	0.03 (0)	0.02 (0)	0.18 (0)

Notes: This table displays descriptive statistics of ideologically polarizing social media boycotts. Panel A shows the most common words used in user profiles for each ideology group that are not used by other groups (word frequencies are in parentheses; stop words and non-English words are excluded). Panel B shows the number of boycotts in each category. Panel C shows descriptive statistics on the type of events triggering boycotts. Panel D shows descriptive statistics on the types of firm responses to boycotts. Panel E shows, by category and boycotter ideology, the average number of posts made when the boycotts trended on X and the day after, as well as average post sentiment. Panel F shows, by user ideology, mean and median user characteristics, with medians italicized and in parentheses.

classification results. Throughout all analyses using ChatGPT, I set the temperature parameter to zero, ensuring that the classification results are replicable and nearly deterministic. After ChatGPT completes the sentiment classification, I draw a random sample of 500 posts and ask a research assistant to independently classify the sentiment directed at the boycotted firm in each post. Comparing model classifications to manual classifications, Table IA2 Panel A of the online appendix shows that GPT-4 has an accuracy (F1 score) of 76% (72%), which exceeds the Twitter RoBERTa-base model's 63% (59%), Loughran-McDonald dictionary's 55% (42%), and FinBERT's 24% (17%).¹⁹ Panel B further shows that GPT-4 consistently performs the best across positive, neutral, and negative posts. These statistics confirm that GPT-4 can better classify entity-specific sentiment of posts than can variants of the BERT model or a dictionary-based approach.

4.3. Descriptive analysis of ideologically polarizing social media boycotts

Table 2 Panel B displays descriptive statistics on the ideologically polarizing boycotts. Eighty (64%) of these boycotts are consistent with liberal beliefs, while 45 (36%) of the boycotts are consistent with conservative beliefs.

Table 2 Panel C examines the events leading to boycotts, summarized through an analysis of news articles and social media posts from the weeks preceding the boycotts. Of the 125 boycotts, 82 resulted from firm decisions grouped into five subcategories, with the most frequent being product or operational decisions, human resources decisions, and political contributions or advocacy. Twenty-six resulted from current or former employee actions or statements, involving both executives and rank-and-file employees. The remaining 17 resulted from external factors, such as politicians' remarks, firms' inaction on polarizing issues, or actions by subsidiaries and franchisees. To examine how firms respond to boycotts, I collected data on firm responses by reviewing news articles and social media posts published in the weeks following each boycott. Panel D shows that the most common response is silence (39%), followed by explanations (33%), reversals of controversial policies (16%) or doubling down (12%).²⁰

Table 2 Panel E shows descriptive statistics on the number of cashtag posts made during polarizing boycotts and their average sentiment. The mean (median) number of cashtag posts when a boycott trends on X is 748 (302). The average user sentiment toward boycotted firms is slightly negative at -0.1 . Among liberal boycotts, those concerning LGBTQ rights (racial equality) have the highest engagement (most negative sentiment), while among conservative boycotts, those focused on politics have both the highest engagement and the most negative sentiment.

Finally, Table 2 Panel F shows descriptive statistics of user characteristics. Of the 45,349 users in my sample, 15,235 (33%) are classified as liberals, 19,745 (44%) as conservatives, and 10,369 (23%) as nonpartisans. The average (median) liberal user follows 53 (16) liberal accounts and six (two) conservative accounts, while the average (median) conservative user follows five (one) liberal accounts and 64 (22) conservative accounts. In terms of financial commentary experience, the average nonpartisan posts 2362 cashtags over eight months, far exceeding the average liberals and conservatives' cashtag count (below 500). This statistic, combined with the fact that nonpartisans are also more likely to have a higher bot score, suggests that nonpartisan users in my sample are more likely to be automated accounts primarily broadcasting investment news or trading signals.

5. Validation test – user ideology and sentiment toward boycotted firms

5.1. Research design

I first validate the conjecture that ideological alignment between social media users and boycotters is associated with sentiment toward the boycotted firm, using the following model:

¹⁹ Accuracy is the proportion of all classifications that are correct. The significant underperformance of FinBERT compared to other methods could be because it was trained on corporate filings, analyst reports, and conference call transcripts (Huang et al., 2023), which differ linguistically from typical social media posts.

²⁰ Untabulated analyses reveal no significant differences in returns between firms that explain, reverse, or double down, compared to those that remain silent.

$$SentTowardsFirm_{i,j,b,t} = \beta_0 + \beta_1 Aligned_{i,j,b,t} + \beta_2 Opposed_{i,j,b,t} + \sum \beta_k UserCharacteristics + \delta BoycottFE + \varepsilon_{i,j,b,t}. \quad (1)$$

The dependent variable $SentTowardsFirm_{i,j,b,t}$ is a categorical variable that can take values -1 (negative), 0 (neutral), or 1 (positive) to indicate social media user i 's sentiment toward firm j during boycott b on date t .²¹ The variable $Aligned_{i,j,b,t}$ is an indicator variable that equals one if user i is ideologically aligned with the boycotters in boycott b targeting firm j and zero otherwise. A social media user is considered aligned with boycotters if the user is a liberal (conservative) and the boycott is consistent with liberal (conservative) beliefs. The variable $Opposed_{i,j,b,t}$ is an indicator that equals one if user i is ideologically opposed to the boycotters in boycott b targeting firm j and zero otherwise. A user is considered opposed to boycotters if the user is a liberal (conservative) and the boycott is consistent with conservative (liberal) beliefs. Note that nonpartisans are neither aligned with nor opposed to boycotters.

Control variables include the following user characteristics that are potentially correlated with sentiment toward boycotted firms: age of the account, decile ranking of cumulative post count in the user's post history, decile rankings of follower and following counts, and verification status.²² Additionally, since users' experience with financial commentary may influence their perceptions of stocks, I control for their financial commenting, measured by the decile rank of users' total cashtag usage in the five months preceding and three months following each post. Finally, because the computer science literature documents that posts made by humans express more extreme sentiment than those by automated accounts (Dickerson et al., 2014), I control for whether the user is likely automated (i.e., "bot"), using the Botometer API to calculate a "bot score" for each user. I classify users with bot scores higher than 4 out of 5 as likely bots.

I cluster standard errors at the firm and user levels. I also include boycott fixed effects, so the regression coefficients are estimated from within-boycott variation. I expect β_1 (β_2) to be negative (positive). That is, compared with nonpartisans, boycotter-aligned (boycotter-opposed) users express more negative (positive) sentiment toward the boycotted firm.

5.2. Descriptive statistics of regression variables

Table 3 Panel A presents the descriptive statistics of regression variables. These statistics are calculated at the post level, meaning that users who contribute multiple posts appear multiple times, so they differ from the user-level statistics in Panel F of Table 2. Of all posts in the sample, 39% are made by users ideologically aligned with boycotters, and 29% are made by those ideologically opposed to boycotters. As of posting date, the average (median) user has been on X for 5.3 (4.5) years, has 14,635 (449) followers, follows 1510 (287) other users, and has made a total of 76,418 (11,782) posts. Users' financial commentary experience is highly skewed: the average (median) user referenced 12,789 (357) cashtags in their posts within an eight-month window. Finally, 3% of posts are from verified accounts and 17% from likely bots.

5.3. Regression results

Table 4 Panel A presents the main results of examining whether ideology affects social media users' opinions about boycotted firms. In Column (1), where the full sample is used, the coefficient on *Aligned* is -0.122 and significant at the 1% level. This suggests that a boycotter-aligned user is more likely than a comparable nonpartisan to post negatively about the boycotted firm. In economic terms, the expected sentiment of a boycotter-aligned user's post is 0.122 points more negative than that of a nonpartisan user's post, equivalent to about a quarter of the standard deviation of nonpartisan post sentiment (0.49). This result is robust to including boycott fixed effects and controlling for other user characteristics, suggesting that it is unlikely driven by contemporaneous firm fundamentals or other observable user characteristics. To further substantiate this finding, I repeat the regression using different subsamples. In Column (2), I present regression results excluding posts containing technical analysis, more than five cashtags, or promotional content, because they are likely to be auto-generated rather than personal opinions. To address concerns that users who are not investment focused drive the main results, I further exclude users with below-median financial commentary experience in Column (3) or those who do not self-describe as investment professionals in their profiles in Column (4). The results under these alternative specifications resemble those in Column (1). Excluding bot accounts from the analyses also yields similar findings (Table 4 Panel B).

Interestingly, in Table 4 Panel A, the coefficient on *Opposed* does not differ significantly from zero in any column, indicating no significant difference in sentiment between boycotter-opposed users and nonpartisans. Boycotter-opposed users' hesitation to support the boycotted firm could stem from a fear of backlash, consistent with the spiral of silence theory (Noelle-Neumann, 1974), which

²¹ Because the dependent variable is ordinal with three possible categories, an ordered logit or ordered probit model is theoretically more appropriate than OLS. However, estimating an ordered logit model with boycott fixed effects through the Stata package *feologit* results in numeric overflow due to the model's computational complexity. Considering Angrist and Pischke's (2009) conclusion that compared with nonlinear models, OLS is easier to implement on panel data and often produces comparable findings, I conduct all my analyses using OLS regression. To confirm the appropriateness of OLS, I estimate an ordered logit model with firm-level control variables from equation (2), user-level controls from equation (1), and no fixed effects through the Stata package *ologit*. The ordered logit regression results yield the same inferences as those in Table 4.

²² Because of the limitation of the Twitter API, I could only access cumulative post count, follower count, following count, and verification status as of the date on which I downloaded the data in January 2023. When constructing regression variables, I use users' January 2023 verification status to proxy for their verification status at the time of post posting. I impute users' cumulative post counts, follower counts, and following counts at the time of the post posting proportional to the ratio of their account age at the time of the post posting to the age of their account as of January 2023. Results in Table 4 remain qualitatively the same if I use the January 2023 values instead of the imputed values as control variables.

Table 3
Descriptive statistics of regression variables.

Panel A: Post-Level Summary Statistics for Regression Variables Used in Table 4						
	N	Mean	SD	p25	p50	p75
<i>SentTowardsFirm</i>	93,447	−0.10	0.55	0.00	0.00	0.00
<i>Aligned</i>	93,447	0.39	0.49	0.00	0.00	1.00
<i>Opposed</i>	93,447	0.29	0.46	0.00	0.00	1.00
<i>UserVerified</i>	93,447	0.03	0.17	0.00	0.00	0.00
<i>AccountAge</i>	93,447	5.30	4.29	1.17	4.51	9.09
<i>FollowerCount</i>	93,447	14,634.75	324,352.30	76.56	449.24	2114.78
<i>FollowingCount</i>	93,447	1509.65	8331.57	45.46	287.32	1104.50
<i>PostCount</i>	93,447	76,418.07	241,540.10	2553.95	11,781.67	50,345.21
<i>FinCommentExp</i>	93,447	12,788.89	45,842.61	19.00	357.00	2931.00
<i>LikelyBot</i>	93,447	0.17	0.38	0.00	0.00	0.00
Panel B: Boycott-Level Summary Statistics for Regression Variables Used in Tables 5–8						
Stats	N	Mean	SD	p25	p50	p75
<i>BHAR(0, +3)</i>	125	−0.59%	4.33%	−2.54%	−0.06%	1.98%
<i>AbTurn(0, +3)</i>	125	0.16%	0.94%	−0.19%	−0.04%	0.25%
<i>AbVol(0, +3)</i>	125	0.05%	1.54%	−0.79%	−0.18%	0.64%
<i>AbIVol(0, +3)</i>	125	2.27%	16.62%	−1.80%	−0.34%	2.63%
<i>AbROIBVol(0, +3)</i>	125	−0.65%	7.78%	−5.82%	0.05%	4.73%
<i>AbROIBTrd(0, +3)</i>	125	−0.29%	8.85%	−4.76%	−0.44%	4.66%
<i>AbRetailTurn(0, +3)</i>	125	0.00%	0.11%	−0.01%	0.00%	0.01%
<i>PctUserAligned</i>	125	34.65%	15.50%	24.00%	33.48%	41.45%
<i>IdeoDiversityScore</i>	125	2.56	0.45	2.34	2.72	2.91
<i>AvgSentTowardsFirm</i>	125	−0.09	0.21	−0.16	−0.02	0.04
<i>BoycotterIdeology</i>	125	0.36	0.48	0.00	0.00	1.00
<i>LogCashtagCount</i>	125	5.55	1.54	4.49	5.71	6.71
<i>LogSize</i>	125	24.58	1.78	23.38	24.42	26.19
<i>BTM</i>	125	0.46	0.73	0.08	0.23	0.53
<i>ROA</i>	125	0.02	0.02	0.00	0.01	0.03
<i>PctInstOwn</i>	125	0.74	0.19	0.64	0.74	0.88
<i>LogNumAnalysts</i>	125	3.14	0.47	2.94	3.18	3.43
<i>AvgTradMediaSent</i>	125	0.00	0.07	−0.01	0.00	0.02
<i>StdTradMediaSent</i>	125	0.06	0.07	0.00	0.05	0.08
<i>LogTradMediaCount</i>	125	2.04	1.38	1.10	2.20	2.89

Notes: This table presents descriptive statistics of regression variables. Panel A presents the summary statistics of variables used in Table 4. *FollowerCount*, *FollowingCount*, *PostCount*, and *FinCommentExp* are displayed in raw numbers instead of decile rankings. Panel B presents the summary statistics of variables used in Tables 5–8. All continuous variables, except for stock returns, are winsorized at the 1st and 99th percentiles.

suggests that individuals are less likely to express their opinions when those opinions are unpopular within a group. Hence, during a polarizing boycott, boycotter-opposed users may feel isolated, due to the widespread outrage on X, leading them to withhold verbal support for the boycotted firms.

Overall, the results in Table 4 indicate that, relative to nonpartisans, social media users ideologically aligned with boycotters express more negative sentiment toward boycotted firms. In contrast, users ideologically opposed to boycotters do not express more positive sentiment relative to nonpartisans, possibly due to a fear of backlash.

5.4. Content analysis of post focuses

Having examined the sentiment of posts and its drivers, I now turn to whether these posts discuss the financial performance of boycotted firms or make political statements. For this task, I use ChatGPT (GPT-4o) to identify whether each post in the sample assesses the financial performance of the boycotted firm and whether it expresses ideological opinions.²³ See Appendix D Panel B for the prompt and Appendix F for example classification results.²⁴

²³ Identifying financial and ideological language requires more judgment than sentiment classification. For this task, I use the GPT-4o model, which outperforms GPT-4 on tasks involving complex language interpretation.

²⁴ After ChatGPT completes the classification, I draw a random sample of 500 posts and ask a research assistant to independently classify the focus of each post. Comparing model classifications to manual classifications, I find that ChatGPT has an accuracy (F1 score) of 79% (78%) when identifying posts referencing boycotted firms' financial performance and an accuracy (F1 score) of 95% (81%) when identifying posts referencing political ideology.

Table 4

User-boycotter ideological alignment and social media user sentiment toward boycotted firms.

Panel A: User-Boycotter Ideological Alignment and Sentiment Toward Boycotted Firms					
Dependent Variable: <i>SentTowardsFirm</i>	Pred. Sign	(1) Full Sample	(2) Excluding technical trends, >5 cashtags, & promotional content	(3) Excluding technical trends, promotional content, >5 cashtags, & low-experience users	(4) Excluding technical trends, >5 cashtags, promotional content, and non-traders
<i>Aligned</i>	(−)	−0.122*** (−3.218)	−0.111*** (−2.783)	−0.089*** (−3.367)	−0.116*** (−4.128)
<i>Opposed</i>	(+)	−0.033 (−0.959)	−0.017 (−0.534)	−0.004 (−0.208)	−0.019 (−0.729)
<i>UserVerified</i>	(?)	0.017 (0.919)	0.019 (1.012)	0.034 (1.348)	0.016 (0.438)
<i>AccountAge</i>	(?)	0.002 (0.952)	0.002 (0.942)	0.003 (1.459)	0.005** (2.442)
<i>FollowerCountRank</i>	(?)	0.005 (1.558)	0.008*** (2.805)	0.006 (1.364)	0.009** (2.320)
<i>FollowingCountRank</i>	(?)	0.002 (0.467)	−0.004 (−1.561)	−0.004 (−1.244)	−0.001 (−0.557)
<i>UserPostCountRank</i>	(?)	−0.017*** (−2.934)	−0.014* (−1.956)	−0.013** (−2.491)	−0.023*** (−4.554)
<i>FinCommentExpRank</i>	(?)	0.018** (2.249)	0.014 (1.233)	0.024*** (3.654)	0.022*** (4.636)
<i>LikelyBot</i>	(?)	0.045** (2.095)	0.086*** (3.229)	0.053*** (2.725)	0.071*** (3.015)
Boycott FE		YES	YES	YES	YES
Clustering		By firm and by user	By firm and by user	By firm and by user	By firm and by user
Observations		93,447	60,882	24,673	16,117
Adj. R-squared		0.184	0.181	0.111	0.126
Panel B: User-Boycotter Ideological Alignment and Sentiment Toward Boycotted Firms (Excluding Users with High Bot Scores)					
Dependent Variable: <i>SentTowardsFirm</i>	Pred. Sign	(1) Full Sample	(2) Excluding technical trends, >5 cashtags, & promotional content	(3) Excluding technical trends, promotional content, >5 cashtags, & low-experience users	(4) Excluding technical trends, >5 cashtags, promotional content, and non-traders
<i>Aligned</i>	(−)	−0.138*** (−2.936)	−0.123*** (−2.502)	−0.115*** (−2.945)	−0.129*** (−3.266)
<i>Opposed</i>	(+)	−0.042 (−1.045)	−0.023 (−0.682)	−0.011 (−0.402)	−0.016 (−0.532)
Controls		YES	YES	YES	YES
Boycott FE		YES	YES	YES	YES
Clustering		By firm and by user	By firm and by user	By firm and by user	By firm and by user
Observations		77,518	53,171	17,662	11,730
Adj. R-squared		0.199	0.185	0.115	0.133

Notes: This table presents results of post-level regressions. Panel A presents regression results of the relationship between user-boycotter ideological alignment and post sentiment. Panel B repeats the test in Panel A excluding users with high bot scores. Decile rankings of *FollowerCount*, *FollowingCount*, *PostCount*, and *FinCommentExp* are used to accommodate data non-normality and outliers. Continuous variables are winsorized at the 1st and 99th percentiles. See Appendix B for detailed variable definitions. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, in one-tailed tests if a directional prediction is specified and two-tailed tests otherwise.

Content analysis suggests that 36% of posts only assess the boycotted firm's financial performance, 6% only make ideological identity-related statements, 6% do both, and 52% do neither.²⁵ In particular, among posts by users ideologically aligned with boycotters, 38% only assess financial performances, 9% only make ideological statements, and 8% do both. These patterns suggest that social media user opinions about boycotted firms could be influenced by considerations related to financial value or identity congruence and that, while financial value posts prevail overall, ideological identity expressions are slightly more common among

²⁵ Posts with no explicit references to either financial performance or political beliefs can still convey investment information. For instance, the following sample post related to Exxon Mobil (\$XOM) reveals the user's share purchase but does not explicitly assess the firm's financial condition: “@DrDividend47 Today I add to the shares I do have \$BAC \$XOM \$EQT \$RBLX” (<https://x.com/clpirtle25/status/1518681123794173952>).

users ideologically aligned with boycotters.

6. Social media user ideology and stock market responses to boycotts

6.1. Descriptive statistics on market reaction to polarizing boycotts

As a starting point for exploring stock market responses to polarizing boycotts, I analyze stock returns around boycotts and their cross-sectional variation. First, in descriptive analyses presented in Panel A of Table 5, there is no significant evidence of negative reaction before the boycotts start trending on X. Panel B shows modest evidence of negative market-adjusted buy-and-hold abnormal returns (*BHARs*) immediately after the boycotts start trending.²⁶ Two-tailed t-tests indicate that the mean *BHAR* is significantly negative at the 10% level for the [0, +1] and [0, +7] day windows. Signed rank tests indicate that median *BHAR* is significantly negative at the 5% level for the [0, +1] day window. Moreover, Fig. 1 displays the mean and 90% confidence interval for buy-and-hold abnormal returns over the [-1, +60] trading days around when a boycott trends on X. The figure shows that polarizing boycotts are associated with an approximate 2.3% decline in equity value from day 0 to day 60, with no sign of reversal in this period, indicating that these boycotts may have a lasting negative effect on equity value.

To understand the cross-sectional variation in market reaction to boycotts, I partition the sample based on whether online users are highly ideologically aligned with the boycotters. High ideological alignment is defined as cases where at least 50% of social media users are ideologically aligned with boycotters. Panel B of Table 5 suggests numerically more negative immediate market reactions to boycotts when online users are highly ideologically aligned with the boycotters, though the differences are not statistically significant in five out of the six specifications.

6.2. Ideology-driven social media opinions and post-boycott returns

To directly examine H1, I estimate the following regression equation:

$$BHAR_b = \beta_0 + \beta_1 PctUserAligned_b + \sum \beta_k Controls + \mu DayofWeekFE + \delta YearFE + \varepsilon_b, \quad (2)$$

where subscript *b* indexes boycotts. The dependent variable *BHAR_b* is the market-adjusted buy-and-hold abnormal return following boycott *b*, measured over three windows. The independent variable *PctUserAligned_b* is the percentage of posts by users ideologically aligned with boycotters out of all posts referencing the boycotted firm's cashtag. Control variables are boycotter ideology, total number of cashtags and news articles (proxy for attention to boycott news), firm size, book-to-market ratio, profitability, analyst following, institutional ownership, average news sentiment (proxies for nature of the news), and standard deviation of news sentiment (proxies for how polarizing the news is).²⁷ I include day-of-the-week and year fixed effects and cluster standard errors by firm.²⁸ I predict β_1 to be negative, consistent with a higher proportion of boycotter-aligned users being associated with more negative market reaction to boycotts.

Table 3 Panel B presents boycott-level descriptive statistics on the regression variables. As of the time of the boycott, an average sample firm has a market value of \$47 billion, a book-to-market ratio of 0.46, and a return on assets of 0.02. Compared with the average Compustat firm-quarter in the same period, which has a market value of \$5 billion, a book-to-market ratio of 2.43, and a return on assets of -0.02, my sample firms have higher market valuations, lower book-to-market ratios, and greater profitability.

Table 6 Panel A reports the results from estimating Equation (2). The coefficient on *PctUserAligned* is negative and statistically significant at the 5% level in Columns (1) and (2) and the 10% level in Column (3) in a one-tailed test, indicating that a higher percentage of opinions coming from boycotter-aligned users is associated with more negative market reaction to boycotts. Panel B indicates that this finding is robust to using the percentile ranking of *BHAR* as the dependent variable instead. This negative relationship suggests that the ideology of social media users can influence market responses to polarizing boycotts.²⁹ Specifically, as the proportion of boycotter-aligned users increases, overall social media sentiment toward the boycotted firms becomes more negative, prompting investors who monitor social media to trade accordingly. This finding demonstrates the capital market implications of ideology-driven social media content.

²⁶ Untabulated analyses show that, out of all 125 boycotts, 70 (56%) had negative *BHAR* over the [0, +1] window, 63 (50%) had negative *BHAR* over the [0, +3] window, and 68 (54%) had negative *BHAR* over the [0, +7] window. None of these proportions differs statistically from 50% based on an exact binomial test.

²⁷ In addition to explicitly controlling for total cashtag counts, untabulated analyses suggest that the results in Tables 6 and 8 are robust to excluding the 12 boycotts that attracted the most attention (highest cashtag counts).

²⁸ To demonstrate that the empirical results are robust to choices of fixed effects, Table IA3 of the online appendix shows that the results in Tables 6 and 8 stay broadly consistent and, in some cases, become more significant without fixed effects, except that, in Table IA3 Panel E Column 2, the coefficient of interest becomes significant at the 10% instead of 5% level.

²⁹ One concern here is that the market reaction to boycotts is confounded by other prior events, such as media articles and earnings announcements. To alleviate this concern, untabulated analyses show that the results in Tables 6 and 8 are robust to controlling for the extent and sentiment of news coverage in the [-7, -1] window. In addition, Figure IA2 of the online appendix plots a histogram of the distribution of boycott trending dates relative to the closest earnings announcement dates. The trending dates do not cluster around or away from earnings announcement dates. Further, only about 6% of boycotts started trending in the [-5, +5] day window around earnings announcements.

Table 5
Mean and median *BHAR* around boycotts.

		(1) Mean	(2) Median
Panel A: Pre-event period			
<i>BHAR</i> (-15, -11)		-0.27%	0.04%
<i>BHAR</i> (-10, -6)		-0.41%	-0.54%
<i>BHAR</i> (-5, -1)		0.46%	0.22%
Panel B: Event period			
<i>BHAR</i> (0, +1)	All	-0.51%*	-0.34%**
	User-boycotter high alignment	-1.59%**	-1.45%**
	User-boycotter low alignment	-0.35%	-0.21%
	High – Low alignment diff.	-1.24%	-1.24%**
<i>BHAR</i> (0, +3)	All	-0.59%	-0.07%
	User-boycotter high alignment	-1.51%	-1.18%
	User-boycotter low alignment	-0.46%	0.29%
	High – Low alignment diff.	-1.05%	-1.47%
<i>BHAR</i> (0, +7)	All	-0.95%*	-0.54%
	User-boycotter high alignment	-1.46%	-2.44%
	User-boycotter low alignment	-0.87%	-0.52%
	High – Low alignment diff.	-0.59%	-1.92%

Notes: This table presents *BHARs* over various windows around the 125 boycotts. Panel A presents mean and median *BHARs* in pre-event windows. Panel B presents mean and median *BHARs* in event windows for all boycotts and by user-boycotter ideological alignment. High (low) alignment is defined as cases where at least (fewer than) 50% of posts are by users ideologically aligned with boycotters. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. For mean *BHARs*, significance is computed based on a two-tailed *t*-test. For median *BHARs*, significance for individual groups is based on a Wilcoxon signed-rank test (against zero), and significance for the high–low alignment differences is based on a two-sample Wilcoxon rank-sum test.



Fig. 1. Post-Boycott *BHAR* and 90% Confidence Interval

Notes: This figure shows the mean and 90% confidence interval of market-adjusted buy-and-hold abnormal returns over the [-1, +60] trading day window around boycott trending dates.

6.3. The mediating role of overall social media sentiment toward boycotted firms

To explore whether the relation between *PctUserAligned* and *BHAR* is mediated by overall social media sentiment toward target firms, in a path analysis, I break down the total effect of *PctUserAligned* on market reaction into direct and indirect effects (Hayes, 2018):

$$\text{AvgSentTowardsFirm}_b = \alpha_1 + \beta_2 \text{PctUserAligned}_b + \sum \theta_k \text{Controls} + \mu_1 \text{DayofWeekFE} + \delta_1 \text{YearFE} + \varepsilon_{b1} \quad (3)$$

$$\text{BHAR}_b = \alpha_2 + \beta_1 \text{PctUserAligned}_b + \gamma \text{AvgSentTowardsFirm}_b + \sum \pi_k \text{Controls} + \mu_2 \text{DayofWeekFE} + \delta_2 \text{YearFE} + \varepsilon_{b2}. \quad (4)$$

Here *AvgSentTowardsFirm* is the average sentiment of posts referencing the boycotted firm's cashtag, measured over the day on

Table 6
Ideology-driven social media opinions and market reaction to boycotts.

Panel A: Ideology-Driven Social Media Opinions and <i>BHAR</i> Following Boycotts				
	Pred. Sign	(1) <i>BHAR</i> (<i>0,+1</i>)	(2) <i>BHAR</i> (<i>0,+3</i>)	(3) <i>BHAR</i> (<i>0,+7</i>)
<i>PctUserAligned</i>	(−)	−0.030** (−1.78)	−0.047** (−2.17)	−0.046* (−1.54)
<i>BoycotterIdeology</i>	(?)	−0.003 (−0.41)	0.002 (0.21)	0.017 (1.11)
<i>LogCashtagCount</i>	(?)	−0.003** (−2.01)	−0.000 (−0.16)	−0.003 (−0.67)
<i>LogSize</i>	(?)	−0.001 (−0.14)	0.001 (0.29)	0.006 (0.85)
<i>BTM</i>	(?)	0.001 (0.09)	0.007 (1.10)	0.006 (0.57)
<i>ROA</i>	(?)	0.102 (1.14)	−0.002 (−0.01)	−0.014 (−0.06)
<i>PctInstOwn</i>	(?)	−0.002 (−0.14)	−0.002 (−0.08)	−0.012 (−0.31)
<i>LogNumAnalysts</i>	(?)	0.008 (0.53)	0.010 (0.44)	0.016 (0.61)
<i>AvgTradMediaSent</i>	(+)	0.081** (1.90)	0.119** (2.17)	0.188*** (2.54)
<i>StdTradMediaSent</i>	(?)	−0.004 (−0.07)	−0.067 (−0.80)	−0.185 (−1.51)
<i>LogTradMediaCount</i>	(?)	0.004 (1.27)	−0.002 (−0.42)	−0.003 (−0.60)
Day of week FE		YES	YES	YES
Year FE		YES	YES	YES
Clustering		By firm	By firm	By firm
Observations		125	125	125
Adj. R-squared		0.065	0.011	0.024
Panel B: Ideology-Driven Social Media Opinions and Percentile Ranking of <i>BHAR</i> Following Boycotts				
	Pred. Sign	(1) <i>BHAR_Rank</i> (<i>0,+1</i>)	(2) <i>BHAR_Rank</i> (<i>0,+3</i>)	(3) <i>BHAR_Rank</i> (<i>0,+7</i>)
<i>PctUserAligned</i>	(−)	−33.327** (−1.90)	−39.348*** (−2.60)	−23.281* (−1.37)
Controls		YES	YES	YES
Day of week FE		YES	YES	YES
Year FE		YES	YES	YES
Clustering		By firm	By firm	By firm
Observations		125	125	125
Adj. R-squared		0.170	0.037	0.008
Panel C: Path Analysis Using Average Social Media Sentiment Toward the Boycotted Firm as the Mediating Variable				
	Pred. Sign	(1) <i>BHAR</i> (<i>0,+1</i>)	(2) <i>BHAR</i> (<i>0,+3</i>)	(3) <i>BHAR</i> (<i>0,+7</i>)
Total Effect	(−)	−0.030** (−1.78)	−0.047** (−2.17)	−0.046* (−1.54)
Direct Effect	(?)	0.001 (0.02)	0.018 (0.37)	0.005 (0.08)
Indirect Effect	(−)	−0.030 (−1.15)	−0.064** (−1.78)	−0.051* (−1.47)
% of Total Effect				
Direct		−2%	−38%	−11%
Indirect		102%	138%	111%
Controls		YES	YES	YES
Day of week FE		YES	YES	YES
Year FE		YES	YES	YES
Observations		125	125	125
Adj. R-squared		0.065	0.011	0.024

Notes: This table presents the regression results of testing Hypothesis 1. Panel A presents the main regression results, Panel B presents the regression results where the dependent variable is percentile ranking of *BHAR*, and Panel C presents the results of the path analysis. Note that in Panel C, t-statistics of the total and direct effects are calculated based on standard errors clustered by firm, whereas t-statistics of the indirect effect are calculated based on bootstrapped standard errors with 5000 replications. See Appendix B for detailed variable definitions. Continuous variables are winsorized at the 1st and 99th percentiles. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively, in a one-tailed test if a directional prediction is specified, and two-tailed test otherwise.

which the boycott first trends on X and the day after. Control variables are the same as those in Table 6 Panel A. β_1 represents the direct effect of *PctUserAligned* on market reaction, $\beta_2\gamma$ represents the indirect effect of *PctUserAligned* on market reaction via *AvgSentTowardsFirm*, and $\beta_1 + \beta_2\gamma$ represents the total effect of *PctUserAligned* on market reaction. I expect the total and indirect effects of *PctUserAligned* on market reaction to boycotts to be negative.

Table 6 Panel C reports the result of estimating this system of equations. (Fig. 2 depicts the results). The total effect of *PctUserAligned* on market reaction and its significance mirror the coefficients on *PctUserAligned* in Panel A. The direct effect of *PctUserAligned* on market reaction is positive and not close to being significant. The indirect effect of *PctUserAligned* on market reaction via *AvgSentTowardsFirm* is negative in all columns and significant at the 5% and 10% levels in Columns (2) and (3), respectively. A percentage breakdown of the total effect shows that it is attributable to the indirect path. Taken together, the path analysis suggests that the negative association between the percentage of boycotter-aligned users and market reaction to boycotts likely operates through average social media sentiment toward the boycotted firm. Importantly, this analysis does not establish causality. Rather, it suggests that more negative overall social media sentiment could be a channel through which ideology-driven social media opinions lead to more negative market reactions to polarizing news.

6.4. Cross-sectional variation in the relation between ideology-driven social media opinions and post-boycott returns

To further mitigate concerns that the association between ideology-driven social media opinions and market response is confounded by other factors, I explore whether this association varies with heterogeneity within social media posts. For a post to influence trading, it needs to be seen and deemed relevant by investors. Thus, I explore whether the visibility and financial relevance of posts moderate the relation between boycotter-aligned opinions and post-boycott returns. To proxy for visibility, I consider a post's total impressions, inclusion of hyperlinks, and repost status. For financial relevance, I consider whether a post's focus is on financial value or political ideology.

First, I test whether the relationship strengthens for high-visibility posts. If ideology-driven opinions move markets, the effect should strengthen in posts more likely viewed by investors, which I first measure directly using each post's impression count (the sum of likes, replies, and reposts). Results in Table 7 Panel A show that the coefficient on *PctUserAligned_PosImp* (posts with at least one like, repost, or reply) is negative and significant at the 5% level in Columns (1) and (2), while the coefficient on *PctUserAligned_ZeroImp* (posts with no like, repost, or reply) is insignificant. Further, I examine two additional factors associated with the visibility of posts: presence of hyperlinks and repost status. Research consistently shows that posts without hyperlinks receive higher engagement, likely because of lower effort required from readers and algorithmic biases against links directing traffic to external websites (e.g., Engelmann et al., 2019; Sehl and Martin, 2021). For reposts, the fact that a post is being reshared implies it is resonating with an audience. Panels B and C show that the association between *PctUserAligned* and market reaction is concentrated among posts without hyperlinks (*PctUserAligned_NoLink*) and reposts (*PctUserAligned_Repost*). Taken with results in Panel A, these results suggest that

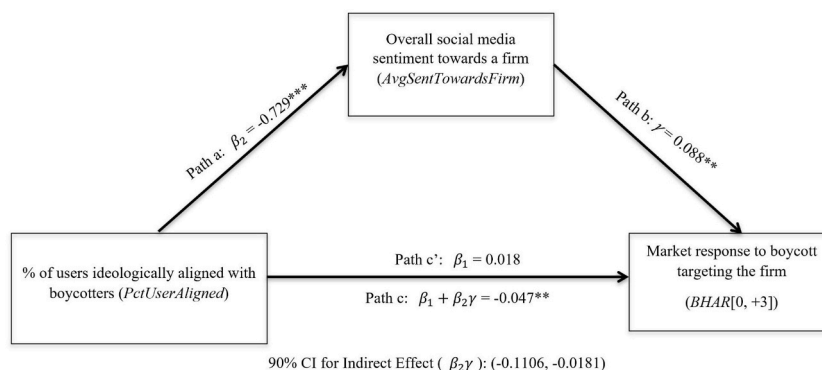


Fig. 2. Path Analysis Diagram – Hypothesis 1 Mediation Analysis

Notes: This figure graphically displays the mediation model used to substantiate Hypothesis 1. For visual simplicity, I only display coefficients estimated with *BHAR* [0, +3] as the dependent variable. Path a represents the effect of *PctUserAligned* on *AvgSentTowardsFirm*. Path b represents the effect of *AvgSentTowardsFirm* on *BHAR* [0, +3]. Path c' represents the direct effect of *PctUserAligned* on *BHAR* [0, +3]. Path c represents the total effect (indirect effect plus direct effect). The t-statistic and one-sided 90% confidence interval for the indirect effect are calculated based on bootstrapped standard errors with 5000 replications. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively, in a one-tailed test when a directional prediction is specified and a two-tailed test otherwise.

Table 7Cross-sectional variation in the relation between ideology-driven social media opinions and *BHAR* following boycotts.

Panel A: Ideology-Driven Social Media Opinions and <i>BHAR</i> Following Boycotts – By Any Post Impression				
	Pred. Sign	(1) <i>BHAR</i> (0,+1)	(2) <i>BHAR</i> (0,+3)	(3) <i>BHAR</i> (0,+7)
<i>PctUserAligned_PosImp</i>	(–)	–0.030** (-1.83)	–0.051** (-2.31)	–0.034 (-1.20)
<i>PctUserAligned_ZeroImp</i>	(–)	–0.064 (-1.14)	–0.005 (-0.04)	–0.148 (-1.12)
Difference	(?)	0.034	–0.046	0.114
Controls		YES	YES	YES
Day of week FE		YES	YES	YES
Year FE		YES	YES	YES
Clustering		By firm	By firm	By firm
Observations		125	125	125
Adj. R-squared		0.062	0.004	0.018
Panel B: Ideology-Driven Social Media Opinions and <i>BHAR</i> Following Boycotts – By Existence of Hyperlinks				
	Pred. Sign	(1) <i>BHAR</i> (0,+1)	(2) <i>BHAR</i> (0,+3)	(3) <i>BHAR</i> (0,+7)
<i>PctUserAligned_NoLink</i>	(–)	–0.089*** (-2.43)	–0.187*** (-4.43)	–0.183*** (-2.78)
<i>PctUserAligned_Link</i>	(–)	–0.011 (-0.47)	0.005 (0.14)	0.003 (0.08)
Difference	(?)	–0.078	–0.192***	–0.186**
Controls		YES	YES	YES
Day of week FE		YES	YES	YES
Year FE		YES	YES	YES
Clustering		By firm	By firm	By firm
Observations		125	125	125
Adj. R-squared		0.078	0.057	0.041
Panel C: Ideology-Driven Social Media Opinions and <i>BHAR</i> Following Boycotts – By Repost Status				
	Pred. Sign	(1) <i>BHAR</i> (0,+1)	(2) <i>BHAR</i> (0,+3)	(3) <i>BHAR</i> (0,+7)
<i>PctUserAligned_Repost</i>	(–)	–0.029** (-1.81)	–0.045** (-2.02)	–0.048** (-1.68)
<i>PctUserAligned_NotRepost</i>	(–)	–0.044 (-0.83)	–0.039 (-0.50)	–0.002 (-0.01)
Difference	(?)	0.015	–0.006	–0.046
Controls		YES	YES	YES
Day of week FE		YES	YES	YES
Year FE		YES	YES	YES
Clustering		By firm	By firm	By firm
Observations		125	125	125
Adj. R-squared		0.058	–0.001	0.014
Panel D: Ideology-Driven Social Media Opinions and <i>BHAR</i> Following Boycotts – By Post Focus				
	Pred. Sign	(1) <i>BHAR</i> (0,+1)	(2) <i>BHAR</i> (0,+3)	(3) <i>BHAR</i> (0,+7)
<i>PctUserAligned_FinOnly</i>	(–)	–0.046* (-1.38)	–0.063 (-1.25)	–0.141** (-2.07)
<i>PctUserAligned_IdeoOnly</i>	(–)	–0.015 (-0.28)	–0.045 (-0.47)	–0.023 (-0.30)
<i>PctUserAligned_Both</i>	(–)	–0.032* (-1.62)	–0.082** (-2.10)	–0.014 (-0.29)
<i>PctUserAligned_Neither</i>	(–)	–0.028 (-0.95)	0.036 (0.68)	0.044 (0.64)

(continued on next page)

Table 7 (continued)

Panel D: Ideology-Driven Social Media Opinions and BHAR Following Boycotts – By Post Focus				
	Pred.	(1)	(2)	(3)
	Sign	BHAR (0,+1)	BHAR(0,+3)	BHAR (0,+7)
Controls		YES	YES	YES
Day of week FE		YES	YES	YES
Year FE		YES	YES	YES
Clustering		By firm	By firm	By firm
Observations		125	125	125
Adj. R-squared		0.044	0.007	0.032

Notes: This table presents regression results examining how various variables moderate the relationship between ideology-driven social media opinions and market reaction to boycotts. Panels A-D present regression results using different partitioning variables: post engagement (A), presence of hyperlinks (B), repost status (C), and post focus (D). See Appendix B for detailed variable definitions. Continuous variables are winsorized at the 1st and 99th percentiles. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, in a one-tailed test if a directional prediction is specified, and a two-tailed test otherwise.

ideology-driven social media posts matter more when they are more visible.

Second, I examine whether the focus of the posts matters, as profit-maximizing investors may find posts with a financial instead of ideological focus more relevant to their trading. Table 7 Panel D shows that posts discussing financial performance, either alone (*PctUserAligned_FinOnly*) or alongside ideological opinions (*PctUserAligned_Both*), are negatively and significantly associated with market reaction to boycotts. In contrast, posts that discuss neither financial performance nor ideological stances (*PctUserAligned_Neither*) or only state political opinions (*PctUserAligned_IdeoOnly*) are never significant. These findings suggest that the relationship between *PctUserAligned* and market reaction is concentrated among posts that are financially relevant, which is more consistent with investors updating beliefs based on economically meaningful information than with behavioral biases driven by ideology.³⁰

Together, these findings suggest that the relationship between ideology-driven social media activity and market reaction to boycotts is concentrated among posts that are more visible and informative to investors. However, because the differences in coefficients have mixed signs and significance, these cross-sectional results are only suggestive evidence of which posts drive the results.

7. Ideological diversity and investor disagreement following polarizing boycotts

In this section, I examine whether post-boycott trading volume and return volatility increase with the diversity of user ideology, as proposed in H2. I estimate the following equation:

$$Outcome_b = \beta_0 + \beta_1 IdeoDiversityScore_b + \sum \beta_k Controls + \mu DayofWeekFE + \delta YearFE + \varepsilon_b, \quad (5)$$

where subscript *b* indexes boycotts. The outcomes I examine include abnormal share turnover (*AbTurn*), abnormal daily return volatility (*AbVola*), and abnormal intraday return volatility (*AbIVola*). The independent variable (*IdeoDiversityScore*), adapted from the country-origin diversity measure of Zhang et al. (2010), is the inverse Herfindahl index of user ideological diversity:

$$IdeoDiversityScore = \frac{1}{PctUserAligned^2 + PctUserNeutral^2 + PctUserOpposed^2}. \quad (6)$$

Because a higher Herfindahl index indicates less diversity, a higher *IdeoDiversityScore* indicates more diversity in users' ideologies. For example, a boycott with an even split of posts by aligned, nonpartisan, and opposed users would have an *IdeoDiversityScore* of 3, higher than that of a boycott with only posts by boycotter-opposed users (*IdeoDiversityScore* = 1). Control variables are the same as those in Equation (2). I also include day-of-the-week and year fixed effects and cluster standard errors by firm. I predict β_1 to be positive, meaning that post-boycott share turnover and return volatility increase with the diversity of user ideology.

Table 8 Panel A presents regression results with average abnormal share turnover over trading days [0, +*t*] as the dependent variable, where day 0 marks the day on which boycotts start to trend on X and *t* equals 1, 3, or 7. Consistent with H2, the coefficient on *IdeoDiversityScore* is positive and significant at the 5% level or better in all columns. Moving from the first to the third quartile of *IdeoDiversityScore* is associated with 0.23% (143% of the mean abnormal turnover) greater four-day abnormal turnover. For control variables, liberal boycotts, more cashtag discussions, smaller firms, less analyst following, more negative or dispersed media sentiment, and more media coverage are associated with higher abnormal turnover.

³⁰ To further distinguish between cash flow and discount rate effects, I analyze the top 100 most frequent words in posts that use financial language. I find that earnings (revenue) is the number 2 (45) most frequent word appearing 2216 (466) times, while none of the top 100 words seem to relate directly to risk or discount rates. I interpret this as consistent with boycott-related social media discourse conveying information about firms' future cash flows while acknowledging that this evidence is suggestive at best and that discount rate could also matter.

Table 8
Ideological diversity and investor disagreement following polarizing boycotts.

Panel A: Ideological Diversity and Abnormal Turnover Following Polarizing Boycotts						
	<u>Pred.</u>	<u>(1)</u>	<u>(2)</u>	<u>(3)</u>		
	Sign	<i>AbTurn</i> (0,+1)	<i>AbTurn</i> (0,+3)	<i>AbTurn</i> (0,+7)		
<i>IdeoDiversityScore</i>	(+)	0.005*** (2.94)	0.004*** (2.58)	0.003** (2.29)		
<i>BoycotterIdeology</i>	(?)	−0.006** (-2.08)	−0.005** (-2.38)	−0.005** (-2.57)		
<i>LogCashtagCount</i>	(+)	0.004*** (3.48)	0.002*** (2.51)	0.001* (1.31)		
<i>LogSize</i>	(?)	−0.005** (-2.64)	−0.002* (-1.86)	−0.001 (-1.09)		
<i>BTM</i>	(?)	−0.004* (-1.71)	−0.002 (-1.02)	−0.001 (-0.79)		
<i>ROA</i>	(?)	0.057 (0.91)	0.027 (0.55)	0.022 (0.45)		
<i>PctInstOwn</i>	(?)	−0.006 (-0.65)	−0.001 (-0.20)	−0.001 (-0.18)		
<i>LogNumAnalysts</i>	(?)	−0.009** (-2.10)	−0.007** (-2.09)	−0.004 (-1.34)		
<i>AvgTradMediaSent</i>	(?)	−0.039** (-2.23)	−0.030** (-2.27)	−0.026** (-2.24)		
<i>StdTradMediaSent</i>	(+)	0.036** (1.81)	0.023* (1.46)	0.020* (1.36)		
<i>LogTradMediaCount</i>	(+)	0.002** (2.15)	0.001*** (2.46)	0.001** (1.76)		
Day of week FE		YES	YES	YES		
Year FE		YES	YES	YES		
Clustering		By firm	By firm	By firm		
Observations		125	125	125		
Adjusted R-squared		0.466	0.321	0.169		
Panel B: Ideological Diversity and Abnormal Daily and Intraday Return Volatility Following Polarizing Boycotts						
	<u>Pred.</u>	<u>(1)</u>	<u>(2)</u>	<u>(3)</u>	<u>(4)</u>	<u>(5)</u>
	Sign	<i>AbVola</i> (0,+3)	<i>AbVola</i> (0,+7)	<i>AbIVola</i> (0,+1)	<i>AbIVola</i> (0,+3)	<i>AbIVola</i> (0,+7)
<i>IdeoDiversityScore</i>	(+)	0.005* (1.58)	0.006** (1.98)	−0.009 (-0.12)	0.028 (0.83)	0.019 (0.94)
Controls		YES	YES	YES	YES	YES
Day of week FE		YES	YES	YES	YES	YES
Year FE		YES	YES	YES	YES	YES
Clustering		By firm	By firm	By firm	By firm	By firm
Observations		125	125	125	125	125
Adjusted R-squared		0.081	0.052	0.040	0.026	0.055

Notes: This table presents the regression results of testing Hypothesis 2. Panel A presents the regression results where the dependent variable is abnormal share turnover, Panel B presents the regression results where the dependent variable is abnormal daily or intraday return volatility. See [Appendix B](#) for detailed variable definitions. Continuous variables are winsorized at the 1st and 99th percentiles. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, in a one-tailed test if a directional prediction is specified, and a two-tailed test otherwise.

In the first two columns of Panel B, where the dependent variable is abnormal daily return volatility, the coefficient on *IdeoDiversityScore* is positive and significant at the 10% level in Column (1) and the 5% level in Column (2).³¹ Moving from the first to the third quartile of *IdeoDiversityScore* is associated with 0.29% (six times the mean abnormal daily return volatility) greater post-boycott four-day abnormal daily return volatility. In Columns (3)–(5), where the dependent variable is abnormal intraday return volatility, the coefficient on *IdeoDiversityScore* is insignificant. This difference could indicate that ideology-driven discourse primarily affects information-based trading rather than high-frequency price fluctuations, which might be more strongly influenced by market microstructure forces.

Taken together, the results in [Table 8](#) suggest that post-boycott trading volume and daily return volatility but not intraday return volatility increase with diversity of social media users' ideologies, partially supporting H2. The link between conflicting ideologies on

³¹ I do not include results for the [0, +1] trading window in Panel B, because it is not particularly meaningful to consider the standard deviation of stock returns over two days.

social media and increased daily return volatility may be a concern for the boycotted firms and their shareholders.

8. Robustness checks and additional analyses

8.1. Endogeneity

Although my findings are consistent with ideology-driven social media opinions influencing market responses to polarizing news, the paper's association-based design raises concerns about omitted correlated variables and reverse causality. I address omitted variable concerns in several ways. First, as noted in Section 6.2, all market-based tests include controls for investor attention (total volume of cashtag posts and number of contemporaneous news articles), the nature of the news (average contemporaneous news sentiment), and how polarizing each boycott is (standard deviation of news sentiment). Second, cross-sectional analyses in Section 6.4 indicate that the association between ideology-driven opinions and market outcome varies predictably with heterogeneity unique to social media posts. Finally, path analysis suggests that ideology-driven social media discourse may influence market reaction via overall social media sentiment toward targeted firms. However, I acknowledge that the existence of other omitted correlated variables cannot be ruled out.

I also take additional steps to address concerns about reverse causality (i.e., market reaction to boycotts affects ideology-driven social media opinions). I instruct ChatGPT (GPT-5.1) to identify posts in my sample that reference the performance of the boycotted firms' stocks prior to the tweets' posting.³² I find that only 7304 (8%) of all 93,447 posts in the sample reference prior stock market performance. When I exclude these posts from the sample, results in Panels A and B of Table 9 suggest that the findings in Table 6 (Panel A) and 8 are robust to this exclusion. Nevertheless, social media users could factor stock performance into their opinions without explicitly mentioning that, so I cannot rule out the possibility of reverse causality either.

8.2. Sensitivity analysis

To test whether the results in Table 4 Panel A are robust to alternative cutoffs of number of political accounts followed, I redefine partisan (nonpartisan) users as those with a high (low) relative ideology difference score, calculated as follows:

$$RelIdeoDiffScore = \frac{|\# \text{ of liberal accounts followed} - \# \text{ of conservative accounts followed}|}{(\# \text{ of liberal accounts followed} + \# \text{ of conservative accounts followed})/2} \quad (7)$$

In Table 9 Panel C, I define liberals (conservatives) as those who follow more liberal (conservative) accounts *and* have a relative ideology difference score above 0.3, effectively requiring higher thresholds for a user to qualify as a liberal or a conservative. Regression results show that the findings in Table 4 Panel A are robust to the alternative cutoff.

8.3. Capital market tests focusing on retail investors

Since boycotts are often started by grassroots individuals or organizations, I examine whether *retail* investor response is associated with ideology-driven social media commentary. Specifically, I test H1 by examining the relation between abnormal retail order imbalance and *PctUserAligned* and H2 by examining the relation between abnormal retail share turnover and *IdeoDiversityScore*. Following the suggestions of Barber et al. (2024), I use the Boehmer et al. (2021) algorithm to identify retail trades and the Lee and Ready (1991) quote midpoint signing method to classify these trades as buys and sells, using code adapted from Blankespoor et al. (2019).

Table 9 Panel D shows the result of testing H1 among retail investors. The association between abnormal retail order imbalance and *PctUserAligned* is consistently negative across all specifications and statistically significant at the 5% level in Columns (2) and (3), where the dependent variable is volume-based abnormal retail order imbalance. Panel E presents the result of testing H2, suggesting that the association between abnormal retail share turnover and *IdeoDiversityScore* is insignificant. Overall, results in Table 9 are consistent with H1 but inconsistent with H2. This might indicate that retail investors only pay attention to social media posts by those ideologically aligned with boycotters. However, caution is warranted when interpreting these findings. First, Barber et al. (2024) and Battalio et al. (2023) document significant misclassifications of retail trades by the Boehmer et al. (2021) algorithm. Second, the power of my tests may be low due to the small sample size.

9. Conclusion

Using 125 polarizing boycotts, this study examines whether ideology-driven social media opinions affect market responses to polarizing firm news. I find that users express more negative sentiment toward the boycotted firm when they are ideologically aligned with boycotters. For market outcomes, the market reaction to boycotts is negative on average and more so when users ideologically aligned with boycotters dominate the social media discourse. This relationship is concentrated among posts that are more likely to be visible and financially relevant. Finally, there is moderate evidence that post-boycott investor disagreement increases with the

³² To improve replicability of the results, I set the model's reasoning effort to none. Comparing manual and ChatGPT classification results in a random sample of 100 posts, I find that ChatGPT has an F1 score of 88%.

Table 9

Robustness checks and additional analyses.

Panel A: Ideology-Driven Social Media Opinions and BHAR Following Boycotts (Excluding Posts Referencing Stock Performance)									
	Pred.	(1)	(2)	(3)					
	Sign	BHAR (0,+1)	BHAR(0,+3)	BHAR (0,+7)					
PctUserAligned	(−)	−0.031** (-1.97)	−0.042** (-2.13)	−0.045* (-1.60)					
Controls		YES	YES	YES					
Day of week FE		YES	YES	YES					
Year FE		YES	YES	YES					
Clustering		By firm	By firm	By firm					
Observations		125	125	125					
Adj. R-squared		0.070	0.008	0.024					
Panel B: Ideological Diversity and Investor Disagreement Following Boycotts (Excluding Posts Referencing Stock Performance)									
	Pred.	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Sign	AbTurn (0,+1)	AbTurn (0,+3)	AbTurn (0,+7)	AbVola (0,+3)	AbVola (0,+7)	AbIVola (0,+1)	AbIVola (0,+3)	AbIVola (0,+7)
IdeoDiversityScore	(+)	0.005*** (2.89)	0.003*** (2.51)	0.003** (2.20)	0.005* (1.65)	0.006** (1.92)	−0.017 (-0.24)	0.028 (0.96)	0.018 (0.94)
Controls		YES	YES	YES	YES	YES	YES	YES	YES
Day of week FE		YES	YES	YES	YES	YES	YES	YES	YES
Year FE		YES	YES	YES	YES	YES	YES	YES	YES
Clustering		By firm	By firm	By firm	By firm	By firm	By firm	By firm	By firm
Observations		125	125	125	125	125	125	125	125
Adjusted R-squared		0.462	0.317	0.165	0.082	0.050	0.041	0.026	0.055
Panel C: User-Boycotter Ideological Alignment and Sentiment Toward Boycotted Firms (Redefining Nonpartisan Users as Those with Relative Following Differences Below 0.3)									
Dependent Variable: SentTowardsFirm	Pred.	(1)	(2)	(3)	(4)				
	Sign	Full Sample	Excluding technical trends, >5 cashtags, & promotional content	Excluding technical trends, promotional content, >5 cashtags, & low-experience users	Excluding technical trends, >5 cashtags, promotional content, and non-traders				
Aligned	(−)	−0.118*** (-3.172)	−0.107*** (-2.654)	−0.088*** (-3.522)	−0.120*** (-4.789)				
Opposed	(+)	−0.025 (-0.810)	−0.011 (-0.371)	0.001 (0.071)	−0.017 (-0.805)				
Controls		YES	YES	YES	YES				
Boycott FE		YES	YES	YES	YES				
Clustering		By firm and by user	By firm and by user	By firm and by user	By firm and by user				
Observations		93,447	60,882	24,655	16,117				
Adjusted R-squared		0.184	0.181	0.111	0.127				
Panel D: Ideology-Driven Social Media Opinions and Retail Order Imbalance Following Polarizing Boycotts									
	Pred.	(1)	(2)	(3)	(4)	(5)	(6)		
	Sign	AbROIBVol(0,+1)	AbROIBVol(0,+3)	AbROIBVol(0,+7)	AbROIBTrd(0,+1)	AbROIBTrd(0,+3)	AbROIBTrd(0,+7)		
PctUserAligned	(−)	−0.055 (-1.12)	−0.077** (-1.71)	−0.059** (-1.86)	−0.058 (-1.02)	−0.040 (-0.82)	−0.027 (-0.61)		
Controls		YES	YES	YES	YES	YES	YES		
Day of week FE		YES	YES	YES	YES	YES	YES		
Year FE		YES	YES	YES	YES	YES	YES		
Clustering		By firm	By firm	By firm	By firm	By firm	By firm		
Observations		125	125	125	125	125	125		
Adj. R-squared		0.108	0.012	−0.067	0.058	−0.031	−0.045		
Panel E: Ideological Diversity and Retail Investor Disagreement Following Polarizing Boycotts									

(continued on next page)

Table 9 (continued)

Panel E: Ideological Diversity and Retail Investor Disagreement Following Polarizing Boycotts				
	Pred.	(1)	(2)	(3)
	Sign	<i>AbRetailTurn</i> (0,+1)	<i>AbRetailTurn</i> (0,+3)	<i>AbRetailTurn</i> (0,+7)
	Pred.	(1)	(2)	(3)
	Sign	<i>AbRetailTurn</i> (0,+1)	<i>AbRetailTurn</i> (0,+3)	<i>AbRetailTurn</i> (0,+7)
<i>IdeoDiversityScore</i>	(+)	−0.000 (−0.06)	0.000 (0.18)	0.001 (0.85)
Controls		YES	YES	YES
Day of week FE		YES	YES	YES
Year FE		YES	YES	YES
Clustering		By firm	By firm	By firm
Observations		125	125	125
Adjusted R-squared		0.399	0.117	0.052

Notes: This table presents additional analyses. Panel A (B) replicates results in Table 6, Panel A (Table 8, Panels A and B) while excluding posts referencing stock performance. Panel C repeats the test in Table 4, Panel A with an alternative definition of nonpartisans (those with relative following differences below 0.3). Panel D tests H1 with abnormal retail order imbalance as the dependent variable. Control variables are the same as those in Table 6. Panel E tests H2 with abnormal retail share turnover as the dependent variable. Control variables are the same as those in Table 8. See Appendix B for detailed variable definitions. Continuous variables are winsorized at the 1st and 99th percentiles. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively, in a one-tailed test if a directional prediction is specified and a two-tailed test otherwise.

diversity of social media users' ideologies. Collectively, the findings show that ideology-driven social media opinions could amplify market responses to polarizing news.

This study is among the first to examine the capital market consequences of ideology-driven social media commentary. In an era where investors increasingly turn to social media for financial information, the findings extend our understanding of how the ideologies of information intermediaries can shape market outcomes, not only through traditional media but also through social media. The evidence suggests that ideology-driven social media commentary around boycotts amplifies price declines and heightens daily return volatility, highlighting the costs social media activities can impose on firms and shareholders in a politically salient crisis.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used ChatGPT in order to assist with copy editing and textual data analysis. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jacceco.2026.101909>.

Appendix B. Complete list of polarizing boycotts in the sample

Category	Issuer Name	Trending Topic
Abortion	C V S Health Corp	#BoycottCVS-08/15/2019
	Lyft Inc	Lyft-09/04/2021
	Meta Platforms Inc	#DeleteFacebook-08/09/2022
Climate change	Walgreens Boots Alliance Inc	Walgreens-07/17/2022
	Exxon Mobil Corp	#ExxonKnew-09/14/2016
		Exxon-10/05/2020
COVID-19	Costco Wholesale Corp	#boycottcostco-05/08/2020
	Delta Air Lines Inc	Delta-04/19/2022
		Delta Air Lines-08/25/2021
	Facebook Inc	Delta CEO-12/28/2021
	Fox Corp	#DeleteFacebook-07/17/2021
		#FoxHatesAmericans-04/27/2021
		#FoxNewsLiedAsPeopleDied-04/04/2020
		#FoxNewsLies-03/18/2020
		Fox's Fake News Contagion-04/01/2020
	GameStop Corp	GameStop-03/19/2020
	Kroger Company	Kroger-12/14/2021
	Live Nation Entertainment Inc	Ticketmaster-11/11/2020
	Pfizer Inc	Pfizer-10/12/2022
	Sinclair Broadcast Group Inc	Sinclair-07/25/2020
	Southwest Airlines Co	Southwest-10/11/2021
	Spotify Technology S A	Spotify-01/25/2022
	Starbucks Corp	#BoycottStarbucks-01/19/2022
	United Airlines Holdings Inc	United Airlines-08/06/2021
		United Airlines-09/29/2021
	Walmart Inc	#BoycottWalmart-01/25/2022
	A T & T Inc	AT&T CEO-05/11/2018
	Activision Blizzard Inc	Blizzard-10/08/2019
	Aetna Inc	Aetna CEO-02/15/2017
	Allstate Corp	Allstate-03/03/2022
	Amazon Com Inc	Whole Foods-08/13/2022
	Coca Cola Co	#BoycottCocaCola-03/04/2022
		#BoycottCocaColaCo-03/27/2021
		#BoycottCoke-03/13/2021
	Comcast Corp	#BoycottNBC-10/14/2020
		Concast-03/20/2020
		MSNBC-04/01/2022
	Darden Restaurants Inc	#BoycottOliveGarden-08/25/2019
	Delta Air Lines Inc	Delta-03/31/2021
	Deutsche Bank A G	Deutsche Bank-03/19/2019
		Deutsche Bank-04/04/2022
		Deutsche Bank-06/29/2018
		Deutsche Bank-08/27/2019
Domestic/international politics	Dine Brands Global Inc	Applebees-02/24/2022
	Disney Walt Co	#BoycottABC-09/13/2019
		#BoycottMulan-08/16/2019
	Exxon Mobil Corp	Hulu-07/26/2022
	Facebook Inc	Exxon CEO-12/10/2016
		#DeleteFacebook-05/05/2021
		#DeleteFacebook-05/25/2019
		#DeleteFacebook-05/28/2020
		#DeleteFacebook-08/28/2020
		#DeleteFacebook-10/14/2019
		#DeleteFacebookNow-06/02/2020
		Facebook-10/14/2020
	Fox Corp	#CancelFox-03/12/2021
		Fox News-06/07/2022
	Goodyear Tire & Rubber Co	Goodyear-08/19/2020
	Grubhub Inc	Grubhub CEO-11/11/2016
	Home Depot Inc	#BoycottHomeDepot-07/08/2019
		#BoycottHomeDepot-10/30/2020
		Home Depot-10/07/2022
	Hyatt Hotels Corp	Hyatt-02/27/2021
	Kellogg Co	#DumpKelloggs-11/30/2016
	Kroger Company	#BoycottKroger-08/04/2021
	Microsoft Corp	Microsoft-01/22/2021
	New York Times Co	#CancelNYT-08/06/2019

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Category	Issuer Name	Trending Topic
Gun policy	Papa John's Intl Inc	#CancelNYT-09/26/2019 Papa John-02/26/2022 Papa John-03/16/2022 Papa Johns-11/06/2019
	PayPal Holdings Inc	PayPal-10/08/2022 PayPal-10/27/2022
	PepsiCo Inc	#BoycottPepsi-01/25/2022
	Sinclair Broadcast Group Inc	Sinclair-04/02/2018 Sinclair-09/13/2019
	Southwest Airlines Co	Southwest-10/30/2021
	Starbucks Corp	#BoycottStarbucks-01/30/2017
	Stellantis NV	Jeep-02/08/2021
	T J X Companies Inc	TJ Maxx-02/08/2017
	Twitter Inc	#TwitterExposed-05/17/2022 #TwitterLockOut-02/21/2018 Twitter-10/14/2020
	Uber Technologies Inc	#BoycottUber-11/11/2019
	Walgreens Boots Alliance Inc	#BoycottWalgreens-02/18/2022
	Walmart Inc	#BoycottWalmart-07/03/2018 Walmart-12/30/2020
	Wayfair Inc	#WayfairWalkout-06/25/2019
	Whirlpool Corp	Whirlpool-08/06/2020
	Amazon Com Inc	#StopNRAmazon-02/22/2018
	Delta Air Lines Inc	Delta-02/24/2018
	Fox Corp	Fox News-05/15/2022
	United Airlines Holdings Inc	United-02/24/2018
	Walmart Inc	#Walmart-09/03/2019 #BoycottWalmart-08/09/2019
	Disney Walt Co	Disney-04/19/2022 Disney-03/30/2022
LGBTQ rights	Exxon Mobil Corp	Exxon-04/23/2022
	Fox Corp	#foxnewsj*****toharstyles-12/30/2020
	Hasbro Inc	Hasbro-02/25/2021
	L Brands Inc	Victoria's Secret-06/16/2021
	Netflix Inc	#NetflixWalkout-10/20/2021 #cancelonnetflix-12/26/2019
	P V H Corp	Calvin Klein-05/12/2022
	Target Corp	Target-11/12/2020
	Yum Brands Inc	Pizza Hut-06/03/2022
	American Airlines Group Inc	American Airlines-09/07/2020 American Airlines-10/25/2017
	Bank Of America Corp	Bank of America-09/01/2022
Racial equality/policing policy	Delta Air Lines Inc	#BoycottDelta-12/21/2016
	FedEx Corp	#BoycottFedEx-05/20/2020
	Fox Corp	#FoxNewsisRacist-06/06/2020
	General Motors Co	Inside the GM-01/17/2019
	Home Depot Inc	Home Depot-03/23/2022
	Nike Inc	#WalkAwayFromNike-07/02/2019 Nike-09/03/2018
	Papa John's Intl Inc	Papa John-07/11/2018 Papa Johns-11/01/2017
	PepsiCo Inc	Pepsi-04/04/2017
	Spotify Technology S A	Spotify CEO-02/07/2022
	Starbucks Corp	#BoycottStarbucks-06/11/2020 #BoycottStarbucks-07/06/2019 #BoycottStarbucks-04/14/2018
	Wells Fargo & Co	Wells Fargo-09/22/2020
	Yelp Inc	Yelp-10/09/2020

Notes: This table lists the 125 boycotts included in the final sample. Profanities are replaced with asterisks.

Appendix C. Variable definitions

Panel A: Variables Used in Table 4		
Variable	Definition	Data Source

(continued on next page)

(continued)

Panel A: Variables Used in Table 4		
Variable	Definition	Data Source
<i>Aligned</i>	A binary variable that equals 1 if the user is ideologically aligned with the prevailing ideological belief of a boycott (i.e., both liberal or both conservative), and equals 0 otherwise.	Twitter Academic API
<i>LikelyBot</i>	A binary variable that equals 1 if one or more of the user's Botometer bot type scores is higher than 4, and equals 0 otherwise. Note that I exclude the Echo-Chamber bot type in this calculation, because users tend to have high Echo-Chamber scores when they are highly ideological and frequently repost political messages. The remaining Botometer bot types are: Fake Follower, Financial, Self-Declared, Spammer, and Other.	Botometer API
<i>Opposed</i>	A binary variable that equals 1 if the user is ideologically opposed to the prevailing ideological belief of a boycott (i.e., one is liberal and the other is conservative), and equals 0 otherwise.	Twitter Academic API
<i>SentTowardsFirm</i>	A categorical variable that can take values -1 (negative), 0 (neutral), or 1 (positive) to indicate a post's sentiment toward a firm targeted by a trending boycott.	OpenAI ChatGPT analysis of posts downloaded through the Twitter Academic API
<i>UserVerified</i>	A binary variable that equals 1 if the user account is verified as of January 2023 and equals 0 otherwise.	Twitter Academic API
<i>AccountAge</i>	The imputed age of a user's X account (in years) as of their post's release.	Twitter Academic API
<i>FollowerCountRank</i>	Decile ranking of a user's imputed follower count as of their post's release.	Twitter Academic API
<i>FollowingCountRank</i>	Decile ranking of a user's imputed following count as of their post's release.	Twitter Academic API
<i>FinCommentExpRank</i>	Decile ranking of cashtag count in post histories spanning five months prior and three months following the post's release.	Twitter Academic API
<i>PostCountRank</i>	Decile ranking of a user's imputed total post count as of their post's release.	Twitter Academic API
Panel B: Variables Used in Tables 5–8		
Variable	Definition	Data Source
<i>AbRetailTurn</i> [0, + <i>t</i>]	Daily average of retail trading volume over trading days [0, + <i>t</i>] divided by total shares outstanding, where <i>t</i> takes the values 1, 3, or 7, minus the firm's average retail share turnover over trading days [-41, -11]. Note that for all abnormal measure calculations, if a firm experienced prior boycotts within the [-41, -11] window, the average (i.e., "normal") measure is instead calculated using the firm's first [-41, -11] window of the calendar year.	TAQ, CRSP
<i>AbROIBVol</i> [0, + <i>t</i>]/ <i>AbROIBTrd</i> [0, + <i>t</i>]	Daily average of retail order imbalance for share volume or number of trades over trading days [0, + <i>t</i>] following boycott trending date, where <i>t</i> takes the values 1, 3, or 7, minus the mean retail order imbalance for share volume or number of trades over trading days [-41, -11]. Retail buys and sells are classified using the Lee and Ready (1991) algorithm. Retail order imbalance measures are calculated following Boehmer et al. (2021) Eqs. (1) and (2).	TAQ
<i>AbTurn</i> [0, + <i>t</i>]	Daily average of trading volume over trading days [0, + <i>t</i>] divided by total shares outstanding, where <i>t</i> takes the values 1, 3, or 7, minus the firm's average share turnover over trading days [-41, -11].	CRSP
<i>AbVola</i> [0, + <i>t</i>]	Standard deviation of raw stock returns over trading days [0, + <i>t</i>] window following boycott trending date, where <i>t</i> takes the values 3 or 7, minus the standard deviation of raw stock returns over trading days [-41, -11].	CRSP
<i>AbIVola</i> [0, + <i>t</i>]	Daily average of intraday return volatility over trading days [0, + <i>t</i>] following boycott trending date, where <i>t</i> takes the values 1, 3, or 7, minus the mean intraday return volatility over trading days [-41, -11].	TAQ
<i>AvgSentTowardsFirm</i>	Average sentiment of posts referencing the boycotted firm's cashtag, measured at the boycott level on day 0 and day +1.	OpenAI ChatGPT analysis of posts downloaded through the Twitter Academic API
<i>AvgTradMediaSent</i>	Average sentiment of all news articles mentioning the boycotted firm on day 0 and day +1. It ranges from -1 to 1, with -1 (1) indicating the most negative (positive) sentiment.	RavenPack Analytics 1.0
<i>BHAR</i> (<i>_Rank</i>) [0, + <i>t</i>]	(Percentile ranking of) market-adjusted buy-and-hold abnormal returns over trading days [0, + <i>t</i>] following boycott trending date, where <i>t</i> takes the values 1, 3, or 7.	CRSP
<i>BTM</i>	The boycotted firm's book value of equity to market value of equity ratio as of the most recent fiscal quarter end.	CRSP and Compustat
<i>BoycotterIdeology</i>	A binary variable that indicates the prevailing ideology of a boycott. 0 indicates that the boycott is consistent with liberal ideological beliefs, whereas 1 indicates that the boycott is consistent with conservative ideological beliefs.	Twitter Academic API
<i>IdeoDiversityScore</i>	One over the sum of the squares of the percentages of posts by three types of users: aligned, opposed, and nonpartisan.	Twitter Academic API
<i>LogCashtagCount</i>	Log of one plus the total number of posts referencing the boycotted firm's cashtag on day 0 and day +1.	Twitter Academic API
<i>LogNumAnalysts</i>	Log of one plus the number of analysts who issued forecasts for the boycotted firm over the most recent fiscal quarter.	I/B/E/S
<i>LogSize</i>	Log of one plus the boycotted firm's market value of equity as of the most recent fiscal quarter end.	CRSP

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Panel B: Variables Used in Tables 5–8		
Variable	Definition	Data Source
<i>LogTradMediaCount</i>	Log of one plus the total number of news articles mentioning the boycotted firm on day 0 and day +1.	RavenPack Analytics 1.0
<i>PctInstOwn</i>	Number of shares owned by institutional investors as a percentage of the target firm's total number of shares outstanding as of the most recent month end.	WRDS SEC Analytics Suite
<i>PctUserAligned(Opposed)</i> <i>[Neutral]</i>	Number of posts by users ideologically aligned with the boycotters (opposed to the boycotters) [neutral], divided by number of posts referencing the boycotted firm's cashtag on day 0 and day +1.	Twitter Academic API
<i>PctUserAligned_Pos(Zero)</i> <i>Imp</i>	Number of posts by users ideologically aligned with the boycotters that have at least one like, reply, or repost (no likes, replies, or reposts), divided by number of posts referencing the boycotted firm's cashtag on day 0 and day +1.	Twitter Academic API
<i>PctUserAligned_ (No)Link</i>	Number of posts by users ideologically aligned with the boycotters that (do not) include hyperlinks, divided by number of posts referencing the boycotted firm's cashtag on day 0 and day +1.	Twitter Academic API
<i>PctUserAligned_ (Not)Repost</i>	Number of posts by users ideologically aligned with the boycotters that are (not) reposts, divided by number of posts referencing the boycotted firm's cashtag on day 0 and day +1.	Twitter Academic API
<i>PctUserAligned_Fin (Ideo)</i> <i>Only</i>	Number of posts by users ideologically aligned with the boycotters that only reference financial performances (ideological beliefs), divided by number of posts referencing the boycotted firm's cashtag on day 0 and day +1.	Twitter Academic API
<i>PctUserAligned_ Both</i> <i>(Neither)</i>	Number of posts by users ideologically aligned with the boycotters that reference both (neither) financial performance and (nor) ideological beliefs, divided by number of posts referencing the boycotted firm's cashtag on day 0 and day +1.	Twitter Academic API
<i>ROA</i>	The boycotted firm's net income divided by its total assets as of the most recent fiscal quarter end.	Compustat
<i>StdTradMediaSent</i>	Standard deviations of sentiment of all news articles mentioning the boycotted firm on day 0 and day +1.	RavenPack Analytics 1.0

Notes: This table provides detailed definitions for variables used in regression analyses.

Appendix D. ChatGPT prompts for textual classifications

Panel A: Classification of Post Sentiment Toward Boycotted Firms

Please answer the following question about the post: What is the sentiment expressed towards {cashtag} in the post? [Positive/Neutral/Negative]

Important instructions:

- Think step by step.
- Explain your answer in a lot of detail, but no more than 5 sentences.
- If the answer is unclear, answer Neutral. The final answer can only be Positive or Neutral or Negative.
- If the returned JSON object contains double quotation marks, make sure to use backslashes to escape them properly.
- Provide answers in the following JSON format:

```
{
  "explanation": <explanation>,
  "final_answer": <Positive/Neutral/Negative>
}
```

The post in question:
{clean_text}
Answer:

Panel B: Classification of Post Focus (Financial Value vs. Ideological Identity)

Please answer the following two questions about the post:

1. Does the post contain language specifically related to the past, current, or future economic, financial, or stock market performance of {cashtag}? [Yes/No]
 - Important instructions:
 - A. Simply mentioning a company's ticker/cashtag or mentioning trading stocks does NOT count as economic/financial language on its own.

- B. Any mention or implication that an event, action, or circumstance could positively or negatively affect the company's performance—including stock price, sales, profits, brand reputation, customer loyalty, future growth, or other outcomes—should be considered “Yes.”
- C. If there is no clear or implied directional impact on the company's performance, answer “No.”
2. Does the post contain language that expresses political or ideological opinions? [Yes/No]
- Important instructions:
 - A. Simply mentioning political events, political figures, or a boycott does NOT count as political or ideological opinion on its own.
 - B. Political or ideological opinions must express the author's views or feelings about political identity, ideology, social or cultural issues, or partisan statements.

Additional Important instructions:

- Consider each aspect carefully.
- Explain your answer with detail, but limit to 5 sentences.
- If the returned JSON object contains double quotation marks, make sure to use backslashes to escape them properly.
- Provide answers exactly in the following JSON format and do not include any text outside this JSON:

```
[
  {\"id\": 1, \"explanation\": <explanation_why_answer_1>, \"answer\": <answer_1>},
  {\"id\": 2, \"explanation\": <explanation_why_answer_2>, \"answer\": <answer_2>}
]
```

The post in question:

{clean_text}

Answer:

Appendix E. Example ChatGPT (GPT-4) classifications of post sentiment toward boycotted firms

Panel A: Trending topic – #BoycottCVS (8/15/2019)

Positive towards \$CVS:

Not sure what all the hub bub is about nor do I care. Must be a bunch of anti vaxxers with flu shots seeing its school time again ... All y'all doing is bringing attetention [sic] to \$CVS. Thanks for the bump in my stocks!#BoycottCVS. (<https://x.com/Coleman4Nh/status/1162050965517873154>)

\$CVS pulling above resistance with nice volume

Crypto dumping in continued months chop following 18 months of 90% losses and almost all alts and 80% loss on Bitcoin. <link> Retrieved from <https://x.com/CryptoLain/status/1162058845126057984>

Neutral towards \$CVS:

Jefferies Financial Group Comments on CVS Health Corp's Q3 2019 Earnings \$CVS <link> #stocks (<https://x.com/AmericanBanking/status/1161956528917753856>)

\$CVS - Current Report Filing (8-k) <link> (<https://x.com/JUIVEDUMAROC/status/1162097156238520321>)

Negative towards \$CVS:

\$CVS #CVSDeniesCare #BoycottCVS is trending and accelerating on X due to denial of coverage for BC pills thru mail order startup NuRx and The Pill Club. Denial likely due to cos apply for coverage as retailer, but really mail order. About to blow up BIG TIME. BEARISH!!! <link> (<https://x.com/larrywabrams/status/1162079384469045248>)

@AngelaBelcamino This is the same board of directors that has driven #CVS into the ground. Won't be too much longer before \$AMZN and \$WMT put \$CVS out of its misery

💩 (<https://x.com/realudaypalled/status/1162081968613445633>)

Panel B: Trending topic – Nike (9/3/2018)

Positive towards \$NKE:

@ClayTravis @DanishaDanielle @Nike Aren't you a free-market capitalist conservative? You can't have it both ways. If the market dictates that Colin Kaepernick is worth that much—more power to him. I know I'll be adding to my \$NKE position and buying more #Nike shoes and apparel because of this. Stay mad Clay. (<https://x.com/AbeFroman/status/1036744818871820288>)

@myhedghog @Nike Polarizing brands, brands with a POV stand out and thrive. Middle of the road consensual brands never do. It's a smart move. \$NKE. (<https://x.com/jfhksar88/status/1036750123609538562>)

Neutral towards \$NKE:

(continued on next page)

(continued)

Panel B: Trending topic – Nike (9/3/2018)

Colin Kaepernick, who is suing NFL owners for allegedly colluding to keep him out of the league, is one of the faces of a new Nike campaign to commemorate the 30th anniversary of the brand's iconic "Just Do It" motto. \$NKE <link> (<https://x.com/SteveKopack/status/1036710414913036289>)
Wooooo buddy cue all the \$NKE posts. (<https://x.com/tycolby/status/1036701591712219137>)

Negative towards \$NKE:

@darrenovell This will go well tomorrow when \$NKE opens up for trading. My bet is this ad will be pulled by Wednesday. Public companies should NEVER get political, never. (<https://x.com/highyield6/status/1036707275065442304>)

Time to short \$NKE? #JustDoIt <link> (<https://x.com/juniorminingpro/status/1036729429605548032>)

Notes: This table displays posts made when "#BoycottCVS" and "Nike" trended on X and their sentiment toward the boycotted firms (CVS and Nike) as detected by GPT-4. "#BoycottCVS" trended on Aug 15, 2019, when liberal users criticized CVS Caremark pharmacy for allegedly lowering reimbursement rates for contraceptives (<https://www.usatoday.com/story/money/2019/08/15/cvs-boycott-pill-club-calls-out-drug-stores-birth-control-rates/2022966001/>). "Nike" trended on Sept 3, 2018, when conservative X users blasted the firm's controversial ad campaign featuring Colin Kaepernick (<https://www.nytimes.com/2018/09/03/sports/kaepernick-nike.html>). Hyperlinks in the posts are represented as <link> for brevity.

Appendix F. Example ChatGPT (GPT-4o) classifications of financial value vs. identity congruence-focused posts (trending topic – grubhub CEO)

Posts focusing on financial value:

insider selling, chart hella bearish, & Uber/\$AMZN soon to be competition? I see \$10 PPS, then BK in your future \$GRUB (<https://x.com/TheWeedSeeker/status/797053003719131136>)

I understand emotions but this is bad for business - half the country (and probably the company) has a different opinion \$GRUB #ShortSale (<https://x.com/HarschCapital/status/797061778710601728>)

Posts focusing on identity congruence:

@M3aloney YOU ARE A RACIST! YOU ARE GARBAGE! SHORT AND SELL \$GRUB (<https://x.com/CoolVideoBro/status/797076026757173248>)

Selling all of my \$grub shares!! That CEO is very intolerant. What a joke! (<https://x.com/bighoots4/status/797076449001881600>)

Posts focusing on financial value and identity congruence:

The "tolerant" left like @M3aloney & his company \$GRUB stock is down 6% today! There is joy in mudville! #BoycottGubHub (<https://x.com/15minofPham/status/797128829391294464>)

Looking forward to shorting this POS. \$GRUB Trades @ 10 x revs w p/e of 70. NOT a Trump economy stock #boycottgrub <link> (<https://x.com/KHerriage/status/796909808637145089>)

Posts focusing on neither financial value nor identity congruence:

\$GRUB covered 34.75 (<https://x.com/mjsoldit/status/797129425582194689>)

\$GRUB Stock Option Volume Spikes 1.9x its Average <link> <link> (<https://x.com/MarketChmln/status/797182491350171648>)

Notes: This table displays posts created when "Grubhub CEO" trended on X and their focuses as detected by GPT-4o. "Grubhub CEO" trended on Nov 11, 2016, when conservatives attacked Grubhub's CEO for sending an anti-Trump email to employees (<https://time.com/4567760/grubhub-ceo-trump-email/>). Hyperlinks in the posts are represented as <link> for brevity.

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